Correlation-Aware Multimodal Analysis of Histopathological Imaging and Genomic Profiles for Breast Cancer

MAJOR PROJECT REPORT submitted in partial fulfillment of the requirements for the Award of the Degree of

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CERTIFICATE

This is to certify that this project report titled **"Correlation-Aware Multimodal Analysis of Histopathological Imaging and Genomic Profiles for Breast Cancer"** is a bonafide record of work done by **Ms. Srinija Naga Swapna Ramisetty (228W1A12I5) and Ms. Gowri Naga Amrutha Samatam (228W1A12I7)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	**Correlation-Aware Multimodal Analysis of Histopathological Imaging and Genomic Profiles for Breast Cancer**
2	**Student Names & Numbers**	**Ms. Srinija Naga Swapna Ramisetty 228W1A12I5** **Ms. Gowri Naga Amrutha Samatam 228W1A12I7**
3	**Name of The Guide**	**Dr. M.Ashok Kumar**
4	**Research Group**	**AI and Genomic**
5	**Application Area**	**Health care**
6	**Aim of the Project**	**To analyse, quantify and correlate histopathologic feature and gene expression data to enhance understanding of breast cancer.**
7	**Project Outcomes**	**1. To develop a standardised multimodal approach to breast cancer research (images and genomics).** **2. To identify correlations between histopathologic morphology and gene expression.** **3. To create an efficient pipeline for feature extraction, reduction of dimensionality and correlation analysis for use in future studies.**

Student Signatures

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Signature of the Guide

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TABLE OF CONTENTS

ACKNOWLEDGEMENT	iii
LIST OF FIGURES	v
LIST OF TABLES	vii
LIST OF EQUATIONS	viii
ABSTRACT	ix
CHAPTER-1 Introduction	1 - 7
1.1Origin of the Problem	2
1.2 Basic definitions and Background	2
1.3Problem Statement with Objectives and Outcomes	5
1.4 Related Applications	6
CHAPTER-2 Review of Literature	8 - 14
2.1Description of Existing Systems	8
2.2Summary of Literature Study	12
CHAPTER-3 Proposed Method	15 - 33
3.1Design Methodology	15
3.2System Architecture Diagram	17
3.3Description of Algorithms	21
3.4 Description of datasets, Requirements and Tools	32
CHAPTER-4 Results and Observations	34 - 58
4.1Stepwise description of Results	34
4.2Test case results/Result Analysis	53
4.3Observations from the work	57
CHAPTER-5 Conclusion and Future work	59 - 61
5.1 Conclusion	59
5.2 Future study	60

References
Appendix

LIST OF FIGURES

Figure **Page**

Figure 3.1 Architecture Diagram of Proposed System 21

Figure 3.2 MLP Architecture 24

Figure 3.3 CNN Architecture 25

Figure 3.4 U-net Architecture 27

Figure 3.5 Routing in Capsule Network 29

Figure 4.1 Before Handling Missing Values 34

Figure 4.2 After Handling Missing Values Using GAIN 35

Figure 4.3 Total features after applying feature reduction 36

Figure 4.4 Raw Dataset Size 36

Figure 4.5 Augmented Dataset Using CTGAN 37

Figure 4.6 Results for Classification using Support Vector Machine (SVM) 38

Figure 4.7 Results for Classification using Multi-Layer Perceptron (MLP) 39

Figure 4.8 MLP Learning Curve 40

Figure 4.9 Results for Classification using Convolution Neural Network (CNN) 41

Figure 4.10 CNN Accuracy Curve 42

Figure 4.11 Results for Classification using Deep Neural Network (DNN) 43

Figure 4.12 DNN Accuracy Curve 43

Figure 4.13 Before and after balancing the dataset 44

Figure 4.14 Representative Augmented Malignant Mammogram Samples 45

Figure 4.15 ROI segmentation in mammograms 45

Figure 4.16 RACapsNet Training and Validation Accuracy 46

Figure 4.17 Masked Input and Grad-CAM Overlay 47

Figure 4.18 Confusion Matrix 48

Figure 4.19 Prediction result with new test image and gene data 48

Figure 4.20 Top Image-Gene Associations 49

Figure 4.21 Correlation Scores for Top Samples 49

Figure 4.22 Top Influential Genes 50

Figure 4.23 Top Gene Expression Values 51

Figure 4.24 Image-Gene Correlation Distribution 52

Figure 4.25 Prediction result with new test interface for image-gene Correlation 52

LIST OF TABLES

Table Name	Page
Table 2.1: Summary of Literature Study	12
Table 4.1: Result Analysis	54

LIST OF EQUATIONS

Equation	Brief Description	Page
Equation 3.1	Rectified Linear unit ranges from 0 to ∞	23
Equation 3.2	Capsule Squashing Function (RACapsnet)	29
Equation 3.3	Dynamic Routing Agreement	30
Equation 3.4	Vector alignment measurement	31
Equation 4.1	Percentage of correct prediction among the total	38
Equation 4.2	Proportion of True Positives among the predicted positives	38
Equation 4.3	Proportion of True Positives among the actual positives	38
Equation 4.4	It is the harmonic mean of precision and recall, providing a single score that balances both metrics.	38

ABSTRACT

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Breast cancer is characterized by complex interactions between tissue morphology and molecular alterations. Understanding these relationships requires effective integration of histopathological imaging and genomic data. This paper presents a correlation-aware multimodal framework for analyzing associations between histopathological image features and genomic expression profiles using breast cancer datasets. Histopathology images are processed using a convolutional neural network to extract high-dimensional visual embeddings representing tissue morphology. In parallel, RNA-Seq gene expression data are normalized and reduced using principal component analysis to capture key genomic variations. To discover cross-modal associations, both feature spaces are projected into a shared latent representation where cosine similarity is applied to quantify correlations between image-derived and gene-derived embeddings. The proposed framework identifies strong cross-modal associations, with several samples exhibiting correlations greater than 0.7 in the latent embedding space. These results demonstrate that multimodal feature representations can capture meaningful relationships between morphological and genomic characteristics of breast cancer. The framework provides a scalable approach for multimodal biomedical data analysis and enables interpretable exploration of associations between visual tissue patterns and molecular signatures. The proposed methodology contributes toward correlation-aware multimodal learning systems that support integrative oncology research and data-driven cancer characterization.

Keywords - Breast Cancer, Histopathological Imaging, Genomic Profiles, Multimodal Data Analysis, Convolutional Neural Networks, Canonical Correlation Analysis, Cosine Similarity, Biomedical Data Integration, Radiogenomics.

CHAPTER - 1

Introduction

1.1 Origin of the Problem

One of the greatest contributors of morbidity and mortality among women around the world is breast cancer [1]. Therefore, it is extremely important to find breast cancer at an early stage so that accurate diagnosis leads to improved patient survival outcomes. Historically, breast cancer diagnosis has relied on independent analysis of either histopathological imaging or analysis of genomic data. Histopathological examination provides visual evidence of tissue structure and morphology and can identify cellular abnormalities [2]. Genomic data analysis can show underlying molecular and genetic changes that occur during cancer progression [3]. When either of these two modalities is used independently from the other for the diagnosis of breast cancer, a complete picture of the disease is difficult to establish because both morphological (appearance) and genetic (Deoxyribonucleic Acid) changes contribute to cancer development.

With advances in medical imaging technologies (e.g., MRI, PET, and CT scans), along with development of high throughput sequencing technologies (e.g., complete genome sequencing), there has been an explosion of heterogeneous biomedical data being generated. Even with these advances in technology, there continue to be significant challenges associated with integrating and analysing disparate data types. Some of the biggest hurdles of integrating and analysing genomic data are its inherent high dimensionality and complexity due to the presence of missing data, noise, duplication of identical features and impossible to eliminate multiple testing consequences of statistically evaluating large numbers of variables. Likewise, extracting meaningful and consistent patterns from digitized histopathological images requires sophisticated deep learning techniques and computational resources [4].

A major barrier to establishing a useful connection between visual data obtained from images and the associated genomic data is the lack of an established structure for effectively combining these two types of information [5]. In reality, opportunities for valuable insight into how tumors behave, develop and are classified may be lost due to this lack of a unified system [8]. Furthermore, traditional machine learning technologies

are often unsuccessful in integrating these two types of data as they are used to develop diagnoses for the problem and produce easily interpretable results and visualizations.

The need to develop a comprehensive multimodal system that will allow for the integration of both histopathological image data and genomic data drove the development of this research project. This research project seeks to add further information and understanding of breast cancer characteristics by using a holistic approach to all aspects of a patient's clinical and diagnostic history; while also aiding in the improvement of the clinician's ability to provide accurate diagnosis and clinical decisions based on the additional information.

1.2 Basic Definitions and Background

1.2.1 Basic Definitions

Histopathologic images: These images are the microscopic images of tissue samples acquired from biopsy methods. These images are primarily reviewed by pathologists in order to find irregularities in the cell structures, such as abnormal shapes, sizes, and patterns that may indicate cancers present in these samples. The histopathologic images contain a large amount of data pertaining to the physical arrangement and orientation of the tissue and the spaces between the cells of the tissue. With the improvement in deep learning with an emphasis on the use of CNNs, histopathologic images can be analyzed automatically to extract data associated with many of the structural and textural characteristics found within the image, which will help with the classification and diagnosis of diseases.

Genomic data: It consists of the level of active gene products within a biological sample (often obtained through the use of microarray methods or RNA sequencing methods). The data collected from these methods will determine whether genes are either switched on or off within a sample of tissue; whether the sample is associated with a cancerous or being associated with tis cancerous and or samples within healthy subject(s). Due to the high-dimensionality of gene expression data where thousands of genes are typically measured, the amount of data associated with an individual sample is substantial, making the analysis of this type of data both time-consuming and requiring a large amount of computational resources. Therefore, in order to extract meaningful biological data from genomic expression datasets, careful processing,

normalization, and feature selection of the data are required prior to proceeding to further implementation of the data into an analysis.

Imputation for Missing Data: Genomic datasets frequently have missing data; often due to technical limitations/errors related to the experiment itself, or in the way that the data were collected. Missing data have substantial effects on machine learning model performance unless they have been properly handled. Accordingly, methods for imputing (i.e. estimating and replacing missing data) are used to ensure that datasets contain complete data and retain important biological relationships. Accurately imputing missing data will increase the accuracy and robustness of any downstream analysis performed on those datasets.

Generative Adversarial Imputation Networks (GAIN): GAIN is a deep learning-based method for imputing missing data using the foundational construct of Generative Adversarial Networks (GANs), which are made up of two neural networks; the 'generator' and the 'discriminator.' The generator's function is to predict the missing data points based on what it has available, while the function of the discriminator is to be able to tell if the predicted missing data points look and behave like the actual data points. Through the process of adversarial training, GAIN is able to impute missing data points that are extremely realistic; this makes it the ideal method for imputing complex types of datasets such as those found in gene expression data.

CTGAN: Generating synthetic data for tabular forms conditional Tabular GAN (CTGAN), a type of deep-learning model, can be used to produce synthetic tabular forms of data; it is extremely important for datasets that contain both structured (e.g., ordered) variables and unstructured (i.e., unordered) variables, as well as imbalanced distribution between different classes. For example, CTGAN could create realistic synthetic samples of gene expression from biomedically relevant datasets that may not contain enough actual observed data to allow for sufficient sample size for confident conclusions. By providing augmented sample sizes to augment insufficiently trained machine-learning models for cancer classification, CTGAN could potentially improve their respective accuracy and generalization capabilities.

Extracting features from data: To create meaningful models from observed data, one must convert raw data into more meaningful representations that are ultimately useful for analysis. In histopathological imaging, convolutional neural networks (CNNs) are

the most widely used method for extracting spatial and texture-based features from images. When analyzing genomic data, feature extraction can either be achieved by some form of statistical analysis (e.g., p-value-based approach) or through the reduction of dimensionality using statistical or dimensionality-related techniques. Most commonly, techniques such as principal component analysis (PCA) are used to reduce the number of features to be considered as candidates for models while maximally preserving relevant data. Success of the feature-extraction process will greatly improve performance of the model produced from it.

Reducing dimensionality of data: One problem with analyzing genomic datasets is that there may potentially be thousands of different gene-level features. This leads many features to become redundant or irrelevant, thus hindering the development of effective gene-based models. Therefore, dimensionality-reduction techniques will reduce the number of gene features but leave those that will add the most to the model produced by analysis. By applying dimensionality-reduction techniques, the computational complexity of creating a gene-focused model can be reduced and improve the accuracy of future models while reducing the likelihood of overfitting the dataset being used to create gene models.

Cosine Similarity: Cosine similarity index uses two dimensions (the angle between the two vectors in n-dimensional space) in order to compute a numerical measurement of how similar two sets of data are to one another. Cosine similarity has a range of values from -1 to +1, where a value closer to +1 indicates a greater degree of similarity between the two data sets than does a value closer to zero. The correlation between histopathological images and genomics by utilizing cosine similarity as a quantitatively determined measure of their level of similarity, can produce a significant amount of information pertaining to the correlations that exist between these two data types (histopathological images and genomic sequences) on a broad and diverse scale.

Multimodal Data Integration: The integration of multimodal data is accomplished by combining data from disparate types of data into one analytical structure in order to obtain a greater and more thorough understanding of complex diseases such as breast cancer through the use of multiple types of data in a complementary fashion. The combination of various types of multimodal data increases the accuracy of predictive capabilities and provides deeper insight than the analysis of individual data modalities.

1.2.2 Background of the Study

Breast cancer is one of the most common and life threatening diseases among women in the world. Early diagnosis and accurate detection may help improve patient survival chances and help plan interventions more effectively. Traditionally, diagnosis of breast cancer is done through histopathological examination by observing abnormal cellular structures and tissue morphology on microscopes [2]. At the same time, the advancement of genomic sequencing technologies has allowed for the study of gene expression patterns that show the molecular mechanisms behind cancer progression and development [3]. Histopathological imaging and genomic analysis have their strengths, but these two medical data types are usually studied separately, limiting the overall understanding of the biological activities of tumors.

New achievements in machine learning and deep learning techniques are now underpinning the analysis of heterogeneous biomedical datasets that allow for feature extraction from both imaging and genomic data [4]. Histopathological images contain structural information of organization tissues, and genomic datasets describe the activity of genes related to cancer. Establishing a correlation between the two modalities can help researchers understand the observable abnormalities in the tissue caused by genetic variants [7]. This way, it will be possible to establish a correlation between morphology and molecular tumor features by measuring the similarity of genomic feature vectors and extracted image features [8].

Therefore, this study grounds on developing multimodal analytical framework that integrates histopathological image features and genomic data to find if they are correlated. The proposed multimodal analytical framework aims to advance the understanding of cancer characteristics and pave the way for future data-driven cancer diagnosis research by employing deep learning–based image feature extraction, reducing the dimensionality of genomic data, and conducting similarity-based correlation analyses.

1.3 Problem Statement with Objectives and Outcomes

1.3.1 Problem Statement

Breast cancer research is focused on the assessment of histopathological images of tissue (images that provide a visual representation of the structure of the tissue) in

combination with using genomic data (data that reflects how genes work). Most studies to date have used the two types of data separately, which limits our understanding of the relationship between the shape and arrangement of the tumor cells and the way that the genes function. Histopathological images provide valuable information about the anatomy of the cancerous tissues and the structure of the cancer cells within that tissue, while genomic data reflect the biological behaviour of the cancer in a more functional sense. The lack of effective means of understanding the association between histopathological and genomic data has created a gap in our understanding of how tissue-level characteristics relate to the molecular activity of breast cancer. Thus, the relationship between histopathological images, and genomic expression patterns must be investigated to improve our understanding of the biological association between the two types of data when examining the biology of breast cancer.

1.3.2 Objectives

- Pre-process histopathological image data and gene expression data obtained from different datasets for multimodal analysis.
- Extract meaningful features from histopathological images and reduce genomic feature dimensions for effective analysis.
- Perform correlation analysis between image features and gene expression patterns and develop a local interface to test and visualize the results.

1.3.3 Outcomes

- A cleaned and structured multimodal dataset prepared from histopathological images and gene expression data.
- Extracted image feature representations and reduced genomic feature vectors suitable for analysis.
- Correlation results indicating the relationship between image features and gene expression patterns, along with a local interface for testing and visualization.

1.4 Societal Applications of the Proposed Work

This proposed framework could greatly influence the field of health care, particularly in regard to the diagnosis and study of cancer. By using a number of sources to define features of diseases, this system will allow for a more detailed perspective of those features. In a clinical setting, this type of framework will allow for

pathologists/researchers to see some features that they might not have been able to see using traditional techniques. Therefore, this will assist with earlier detection of cancer, increase accuracy of diagnosis, and assist with treatment planning.

Additionally, the work will allow for advancements in the concept of tailored medicine by providing treatment plans to patients based on their genetic and morphologic profiles. Furthermore, the proposed work will support current research efforts to identify novel biomarkers as well as understand how cancer progresses. Thus, this work encourages the utilization of advanced computer techniques in the practice of health care, resulting in better care for patients and more efficient systems for the analysis of medical data.

CHAPTER-2

Review of Literature

2.1 Description of Existing Systems

This work attempts to study histopathological image features and their relation to the molecular data in breast cancer through digital pathology image and transcriptomic-proteomic data integration from TCGA and CPTAC in the study "Correlation Analysis of Histopathology and Proteogenomics Data for Breast Cancer" by Xiaohui Zhan et al. [1]. An automated computational pipeline was deployed for the extraction of quantitative features of morphological features based on nuclear size, shape, and spatial distribution from the tissue image. These parameters were then correlated with gene expression and protein abundance, to analyze the molecular basis of tumor morphology. The groups of biological processes strongly associated with the morphological characteristics include metabolism, immune response, cell cycle, and extracellular matrix-related development. Several morphological features were also associated with patient survival. Thus this study emphasizes the value of histopathological image and multi-omics integration in breast cancer.

The main idea of this work is "The key is to harness histopathological image features associated gene expression patterns to predict molecular characteristics and prognosis of head and neck squamous cell carcinoma (HNSCC)" in the paper "Histopathological Image and Gene Expression Pattern Analysis for Predicting Molecular Features and Prognosis of Head and Neck Squamous Cell Carcinoma" by Linyan Chen et al. [2]. Quantitative morphological characterization of tumor tissues was performed using digital pathology images and genomic data from TCGA. Different machine learning models were used for the classification of somatic mutations and molecular subtypes based on the image features. The correlation of image-derived features and gene expression profiles were further explored using bioinformatics analysis. Finally, a prognostic model was proposed by combining histopathological image features, termed the integrated transcriptomic phenotype of HNSCC for survival prediction. The findings identify the potential of combined pathology image analysis with genomic data for cancer prognosis evaluation.

This study gives a full molecular characterization of the Breast cancer histologic types based on The Cancer Genome Atlas (TCGA) data in the research "Molecular Analysis of TCGA Breast Cancer Histologic Types" by Aatish Thennavan et al. [3]. In the analysis, the authors relied on more than 1,000 samples of breast cancer and supplemented the data with extensive histological annotations performed by qualified pathologists. Transcriptomic and genomic analyses were performed for six rare histologic types, including cribriform, micropapillary, mucinous, papillary, metaplastic, and medullary carcinomas. Molecular signature was identified through differential gene expression (DEG) and pathway enrichment analysis for each histologic type. Histological features were also integrated with genomic data to classify tumors into biologically relevant groups. Specific gene signatures were also identified that could discriminate certain histologic types in multiple datasets. In sum, the authors show that combination of histological and genomic data can add to the understanding of the heterogeneities of breast cancer histological types and enhance tumor classification.

The study suggests a deep learning framework called hist2RNA to directly predict gene expression from breast cancer histopathology images in the paper "hist2RNA: An Efficient Deep Learning Architecture to Predict Gene Expression from Breast Cancer Histopathology Images" by Raktim Kumar Mondol et al. [4]. The pretrained convolutional neural network extracts visual feature vectors from patches of images. These patient-level aggregated features are then passed through convolutional layers to predict expression levels for 138 genes, which are associated with molecular profile of breast cancer. The final gene expression prediction is also used in molecular subtype classification, including Luminal A and Luminal B, and prognostic analysis. It shows molecular activity can be deduced from the histopathology image morphological motifs. The study proves that deep learning models can be a speedy and affordable alternative to the conventional molecular testing methods for gene expression analysis.

For this ImageCCA approach, the goal is to analyze the relationship between histopathological image features and gene expression levels in the work "Joint Analysis of Expression Levels and Histological Images Identifies Genes Associated with Tissue Morphology" by Jordan T. Ash et al. [5]. The proposed framework, called ImageCCA, analyses relations between histopathological image features and gene expression levels. The framework has been tested on datasets from TCGA breast

cancer, lower grade glioma (LGG), and GTEx projects with paired histological image and RNA-Seq gene expression profiles. These subsets also were correlated with morphological features of extracellular matrix, neuronal, and muscle tissues. Finally, genotype–image feature associations that account for phenotype variation in tissue morphology across samples were discovered. In general, then, the work highlights the potential for the analysis of histological images in conjunction with RNA-Seq gene expression data to discover biologically significant associations between microanatomical features and genomic factors.

The overall aim of this radiogenomics study is to probe the gene expression pattern in breast cancer and identify their association with MRI-derived radiomic features and map the findings accordingly in the study "Radiogenomics of Breast Cancer Using Dynamic Contrast Enhanced MRI and Gene Expression Profiling" by Albert C. Yeh et al. [6]. MRI imaging was obtained using dynamic contrast-enhanced (DCE)-MRI, and gene expression results were obtained using RNA sequencing (RNA-Seq) from tumor tissue samples. In total, 38 radiomic features characterizing tumor size, shape, texture, morphological, and enhancement-related characteristics were generated from MRI images. Gene Set Enrichment Analysis (GSEA) was used to identify the associations of radiomic features in pathways of biological genes (186 pathways). Tumor size and shape features were positively associated with cell proliferation pathways and negatively associated with apoptosis pathways, while immune-related pathways also showed strong correlations with imaging features. The study also demonstrates the possibility of linking the radiomic features from MRI to gene expression status, hence supporting the non-invasive interpretation of tumor biology based on image-derived phenotypes in breast cancer.

This study proposed a multi-omics network-based stratification (Multi-NBS) method that combined somatic mutation data and RNA-Seq gene expression data to improve cancer subtype identification in the research "Integrating Genetic and Gene Expression Data in Network-Based Stratification Analysis of Cancers" by Kenny Liou and Ji-Ping Wang [7]. The approach is to perform a joint analysis of two different types of genomic data which are somatic mutation data and RNA-Seq gene expression data into an integrated molecular profile mapped onto a gene interaction network. Network propagation and network-regularized non-negative matrix factorization (NMF) are used next to cluster the patients into biologically sound tumor subtypes. The TCGA

datasets were for ovarian, bladder, and uterine cancer cases to analyze the subtype properties and clinical relevance of the framework. This integrated method was able to detect subtype-specific genes and biological pathways signature to tumor progression and survival outcomes. In sum, the study shows that integrating multiple genomic data types in a network-based framework improves the stratification in cancers and enhances the knowledge of tumor biology.

The present study aims at proposing a deep learning framework, Genome-informed Hyper-Attention Network (G-HANet) to enhance cancer survival prediction based on histopathology whole slide images (WSIs) and genomic data in the paper "Histo-Genomic Knowledge Association for Cancer Prognosis from Histopathology Whole Slide Images" by Zhikang Wang et al. [8]. In this, training data necessitate gene expression profiles, and prediction-task-based correspondences between genes and image features are iteratively optimized. These associations are then exploited for survival modeling and prediction through a hyper-attention survival branch. This enables genomic knowledge to be distilled into image-based models such that only WSIs may be needed in the inference stage. To do so, the framework was evaluated on five TCGA cancer datasets with breast and lung cancer patients to analyze prognosis performance. This work shows that histo-genomic knowledge is capable of improving the performance of WSI-based models for cancer prognosis analysis.

This study performs a radiogenomic study on breast cancer immune-related gene expression and clinical and ultrasonic imaging features in the work "Radiogenomic Analysis of Clinical and Ultrasonic Characteristics in Correlation to Immune-Related Genes in Breast Cancer" by Tingyao Dou et al. [9]. Data on gene expression were obtained from the GEO and IMMPORT databases, and differentially expressed immune-related genes associated with patient survival were screened using bioinformatics analysis. Relevant clinical information, ultrasonic imaging characteristics, and tumor tissue specimens from patients with breast cancer were collected for whole transcriptome sequencing and immunohistochemical validation. Eight immune-related genes were associated with OS, including CXCL2, MIA, NR3C2, PTX3, S100B, SAA1, SAA2, and CXCL9. The results of the correlation analysis indicated that there are several ultrasonic characteristics and clinical factors such as; The results prove that an ultrasonic imaging feature based radiogenomic

approach can be used as a functional tool in breast cancer diagnosis, prognosis assessment and immunotherapy studies.

At last, This paper briefs on the use of artificial intelligence (AI) in digital pathology for neoadjuvant therapy (NAT) response prediction in breast cancer patients in the review "Artificial Intelligence-Based Pathology to Assist Prediction of Neoadjuvant Therapy Responses for Breast Cancer" by Juan Ji et al. [10]. There is also a discussion about the type of morphological feature extracted by AI models from histopathological images, such as nuclear pleomorphism (indicates the variability of the size and shape of tumor cell nuclei), mitotic activity, stromal characteristics (desmoplastic, normal, or fibrotic), and tumor-infiltrating lymphocytes (TILs). Convolutional neural networks (CNNs) are some of the widely in machine learning and deep learning techniques for the automatic recognition of complex tissue motifs for diagnosis and prognosis evaluation of pathologists. The review also covers histopathology, radiology, genomics, and clinical data incorporation as multi-modal data sources to improve prediction accuracy. Further, challenges involving dataset(s) limitations, model interpretability, and clinical standardized validation needs are elaborated on. AI-based pathology has significant potential for personalized treatment planning and therapy outcome prediction for breast cancer patients.

2.2 Summary of Literature Study

Table 2.1: Summary of Literature Review

Author Name, Year, Cite No.	Paper Title	Model / Methods used	Work Done	Conclusion
Xiaohui Zhan et al., 2019 [1]	Correlation Analysis of Histopathology and Proteogenomics Data for Breast Cancer	Statistical correlation analysis	Image features were extracted from histopathological images and correlated with transcriptomic and proteomic data from TCGA and CPTAC.	Mid-level correlation tissue morphology to molecular pathways.

Linyan Chen et al., 2021 [2]	Histopathological Image and Gene Expression Pattern Analysis for Predicting Molecular Features and Prognosis of HNSCC	CNN, correlation techniques	They extracted morphological information from images produced by pathology and associated those features with gene expression profiles.	Image features and genomic data – moderate association.
Raktim Kumar Mondol et al., 2023 [4]	hist2RNA: An Efficient Deep Learning Architecture to Predict Gene Expression from Breast Cancer Histopathology Images.	Deep learning CNN model	Apart from the predictions of gene expressions and molecular subtypes, visual features were also extracted from the histopathology images.	Had a moderate ability to predict between image morphology and gene expression.
Jordan T. Ash et al., 2021 [5]	Joint Analysis of Expression Levels and Histological Images Identifies Genes Associated with Tissue Morphology	Convolutional Autoencoder + CCA	Performed association study between histology image features and the RNA-Seq gene expression.	Moderate cross-modal correlation is observed.
Albert C. Yeh et al., 2019 [6]	Radiogenomics of Breast Cancer Using DCE-MRI and Gene Expression Profiling	Radiomics + GSEA	MRI radiomic features were extracted and associated with gene expression pathway analysis.	Imaging features were found to have a moderate correlation with biological pathways.

Tingyao Dou et al., 2025 [9]	Radiogenomic Analysis of Clinical and Ultrasonic Characteristics in Correlation to Immune-Related Genes in Breast Cancer	Radiogenomic correlation analysis	Correlated the features of ultrasound images with the patterns of gene expression related to immunity.	Moderate correlation between imaging feature and genetic

The literature study in Table 2.1 presents the most relevant research works on the medical imaging features/genomic or molecular data associations analysis for cancer. On this premise, several studies have tried to combine single/ multi-modal medical imaging data with genomic/ molecular level information to characterize tumor biology and provide prognosis more accurately. For example, Zhan Xiaohui et al. analyzed the correlations between histopathological morphology and proteogenomic data, and Chen et al. conducted a study of the associations between histopathological image features and gene expression patterns to predict molecular characteristics. Ash et al., also proposed an ImageCCA framework for the discovery of associations between morphology and gene expression profiles, and Yeh et al. also conducted a radiogenomics study of links between radiomic features from MR imaging with genomic pathways. Mondol et al. proposed the hist2RNA deep learning framework for the prediction of gene expressions based on histopathology images while Dou et al. studied the association between ultrasound imaging features and immune-related gene expressions for breast cancer.

Most do present some level of association between the attributes derived from the imaging process and gene expression profiling, but fail to obtain high levels of correlation. Most of the existing works report show a moderate relationship between imaging features and genomic information. Hence, a mutual relationship is to be established/improved by correlating histopathological image features and genomic information with a cosine similarity of 0.7 for the proposed method. Therefore, this is considered as a measure of efficiency (or otherwise) of the considered multimodal analysis framework.

CHAPTER-3

Proposed Method

3.1 Design Methodology

3.1.1 Multimodal data preparation

The first step in the proposed methodology involves the creation of a multimodal breast cancer dataset that consists of histopathology images and genomic expression data [1]. The mammogram images will be derived from a comprehensive digital pathology resource that contains millions of slices of breast tissue images sorted by their associated tumor characteristics. Genomic information will be obtained from the TCGA-BRCA dataset for the gene expression levels of RNA-Seq [3]. These datasets are complemented by their respective biological characteristic (histological image versus RNA expression), where histological images provide a morphological representation of cells at the tissue level, while RNA expression provides a molecular representation of the alteration to the cellular. In preparation for analysis, both datasets will be converted to structured matrices and will be aligned in terms of sample size to maximise the possibility of a joint computational process.

3.1.2 Histopathological Image Feature Extraction

The extraction of features from histopathological images occurs after preprocessing the images to establish consistency in resolution and format for the extraction process using a deep convolutional neural network. Image preprocessing includes resizing each tissue patch to a fixed resolution, normalizing each tissue patch into a tensor representation, and preparing tissue patches for inference by a neural network. After all of these preprocessing steps, a deep convolutional neural network architecture is applied to extract high-dimensional visual embeddings for the histological specimen, which represent the morphology of the histological specimen. A deep convolutional neural network learns multiple hierarchical feature representations that describe the structural features of the histological specimen, including the organization of cells, texture characteristics and structures associated with the tumor micro environment [8]. The final output from the convolutional neural network is a compact representation of the histological specimen that summarizes its visual characteristics.

3.1.3 Genomic Data Normalization and Reduction of Dimensions

Researchers experience significant challenges in integrating high-dimensional genomic data (e.g., gene expression data) into imaging features using computational techniques due to the number of gene expression values contained within a typical genomic dataset (thousands of gene expression values) [9]. Therefore, each gene in the gene expression data is normalized by applying a standardization process to create equal scaling across genes for sample(s) processed with the same method. This process of standardization provides assurance that gene expression values generated from each gene will not bias the analysis conducted later on the samples analyzed. After the gene expression data has been standardized, dimensionality reduction is performed on the high-dimensional gene expression matrix to create low-dimensional representations of the gene expression matrix while still retaining the maximum amount of informative variance within the gene expression dataset. Principal component analysis is used to extract the principal components of the gene expression dataset, representing the genes with the highest amount of variance.

3.1.4 Image Feature Embedding Transformation

Further processing of the high dimensional convolutional neural network generated embeddings resulting in a compact representation that can be used in multimodal analysis. Image feature vectors undergo dimensionality reduction in order to eliminate duplicate data and retain the necessary structural patterns as determined by the neural network. Principal Component Analysis transforms the CNN feature space into a lower-dimensional embedding space while still capturing the variance associated to morphological tissue characteristics. This process allows the comparison of the CNN derived feature space with the gene-derived feature space in a more efficient manner.

3.1.5 Latent Multimodal Representation Construction

Compact Image Embeddings and Genomic Embeddings both then project into a common latent representation space [5]. Standardized feature vectors for both modalities will be aligned by scaling/examining variance at this point to ensure the set of features are comparable. The resulting latent representation for both modalities contains the most important features of each while allowing for a cross-modal comparison. This combined feature space will allow for the identification of

relationships between morphological tissue characteristics and signature gene expression levels.

3.1.6 Correlation Ranking and Association Identification

The similarity scores from the cross-modality analysis across all samples were used to determine the strongest multimodal associations. The correlation values were ranked in descending order from the highest correlational values to lowest to emphasize: the strongest multimodal relationships exhibited between the image features of tissues and the genomic feature of expression patterns of genes. The samples with the highest correlation values (i.e. greatest similarity) indicate the strongest associations between the morphological characteristics of tissues and the gene expression patterns in the same sample. The data for these samples will then be analyzed in detail and interpreted to determine the degree to which their morphological and genomic characteristics represent the tissue of origin.

3.1.7 Visual and Interpretation of Multimodal Associations

The last segment of the methodology is focused on visual presentation and interpretation of the discovered multimodal associations. Statistical plot-type graphics are used to display the correlation distributions from both modality to illustrate the overall relationship of tissue morphology and gene expression. Using these graphical representations of the correlation distributions in conjunction with a selection of representative histopathologic images associated with the highly correlated samples to allow qualitative interpretation of tissue morphology. With this visual presentation framework, researchers can investigate and validate the relationship that exists between visual patterns in the tissue and genomic expression characteristics of the breast cancer dataset [8].

3.2 System Architecture Diagram

The architecture depicted in Figure 3.1 shows what a complete workflow for breast cancer analysis would look like by integrating histopathological imaging and genomic data into a multimodal framework. The workflow consists of data collection/preparation, feature extraction, dimensionality reduction, and classification/correlation analysis within a single pipeline enabling robust accuracy and interpretability of results. The workflow has four main areas of focus; data preparation,

feature processing, multimodal integration, and analysis/visualization. During data preparation, histopathological imaging and genomic datasets are collected and pre-processed using normalization and cleaning techniques to create a consistent dataset. During feature processing, convolutional neural networks (CNN) are utilized in CNNs to extract deep features from the histopathological images [4], while principal component analysis (PCA) is performed on the genomic data to reduce dimensionality while retaining the greatest amount of useful information. During multimodal integration, cosine similarity is applied to the extracted image feature representations and the converted genomic feature vectors to find similarities between the structural and molecular features of breast tissue. Following multimodal integration, the result of the similarity calculations is aggregated and displayed in a user-friendly manner for ease of interpretation/testing of new data; consequently, a complete and integrated understanding of the characteristics of breast cancer can be established. The process is described as follows:

1. **Data Preparation:**

This step outlines the initial phases of the proposed architectural framework, including the collection of data on breast cancer patients through two separate modalities (microscopic histopathological examination of tissue samples; and RNA-Seq gene expression profiling using the TCGA BRCA application programming interface (API)). The histopathological data consists of thousands of independent histological samples that have been obtained and compiled into a single, large digital pathology database of breast cancer tissue specimens and made available over the Internet (rather than through an API) [2]. The genomic data consists of many RNA-Seq-based expression profiles for breast cancer patients that are available to researchers through an API from the TCGA BRCA. The two datasets have been structured such that they can be easily processed using various computational tools for the pre-processing and extraction of relevant attributes prior to beginning the learning phase of our proposed system.

2. **Histo-pathology image processing & features extraction:**

Histopathological images will be pre-processed in several different ways prior to being able to use these images for machine learning with Deep Learning through Convolutional Neural Networks (CNNS). Preprocessing involves resizing and

radiometrically normalizing the images and performing tensor transforms on the images. Several different high dimensional feature embeddings will be produced that are representative of the different areas of interest (e.g., cellular patterns, e.g. cell types and/or tumours; and the different spatial arrangements of the cells) in a histopathological image as a result of the use of CNNs. More specifically, there will be a high dimensional embedding that contains learned features from the most prominent morphological feature identified in a histopathological image. All of the embeddings will ultimately provide a numeric representation of the histopathological features present in the tissue and will also be made available as part of an integrated data set for future use as a complement to genomics data [4].

3. Genomic Data Processing & Feature Reduction:

Theoretically, the genome sequence data are prepared to perform a profile analysis of multiple measurements starts from the most time-consuming, the RNA-Seq, from which the required HG (high-dimensional) matrices (in this case, RNA expression data) were generated. The first step in preparing genomic data to further be processed and have dimensionality reduced is to first normalize (or standardize) the individual gene expression measurement values to adjust for differences in scale and variance associated with individual gene expression data. Several methods of dimensionality reduction can be applied to HG genomic databases, allowing the user to take a large number of dimensions of RNA expression (m) matrix and compress them into relatively small (latent) information that represents RNA expression profiles. The Principal Component Analysis (PCA) identifies and/or supports the largest or most relevant sources of variance in raw (unfiltered) RNA expression data while also minimizing the associated amount of noise or redundancy (caused by dummy measures) associated with the data [10].

4. Multimodal Latent Representation Learning:

In order to perform a cross-modality comparison between two different modalities (images and genomics), both modalities will be generated from a latent representation that represents both sets of extracted, processed and embedded features in a common latent representation space after both sets have undergone a feature extraction, processing and embedding process. The alignment of both sets of features within this common latent representation space will occur through the use of feature information

from both image and genomic modalities, so as to identify any features that overlap in both modalities by utilising multimodal representation learning techniques for dimensionality reduction, thereby providing a shared feature space in which the relationships between the visual feature of the tissue and the corresponding expression feature for the genes can be compared. The end result of this process is the ability to locate and compare representations of identical tissue formed from a single image and one derived from the gene expression of that tissue but constructed in the same latent space, which provides a reliable and interpretable model by including the rationale behind the model's output [11].

5. Correlation-Based Multimodal Analysis:

The relationship (correlation) between the various features of images (extracted from histopathological images) and the gene expression features (extracted from gene expression data) will be analyzed using a combination of Multiple Correspondence Analysis on the features across both image (e.g. pixel values) and gene expression (e.g. the CLC homology identified) and the development of a common latent representation for both the image feature set and the gene feature set. Various similarity metrics (will be) used to determine the degree of similarity of the image features and gene features based upon their representations in a common latent space (i.e. the feature vectors of images and gene expression are similar to each other). Cosine similarity will be used to indicate the strength of the correlation between the feature vectors of the histopathological image features and the feature vectors of the gene expression features in that they are part of the common latent space created previously. Therefore, a large value for cosine similarity will indicate a strong correlation between the image features and gene features, providing evidence to support the strong correlation between images and gene expression.

6. Graphical Visualization and Correlation Aggregation Interface:

The last phase in the development of the system architecture includes the collection of multimodal correlations and the visual representation of those correlations. The order of the similarity scores produced from the analysis of these correlations creates a visual representation of the strength of relationship (similarity) between particular histopathological images and the genomic profiles that correlate with those images. Also displayed will be the distributions of the correlations as well as the sample

locations that have a very high strength of correlation, which will lead to the identification and the visual display of the corresponding histopathological image and will lead to an intuitive evaluation of the relationship between a particular cancer's histopathology or cancer's genetic profile along with potentially prompting future investigations related to the multimodal characteristics of cancer.

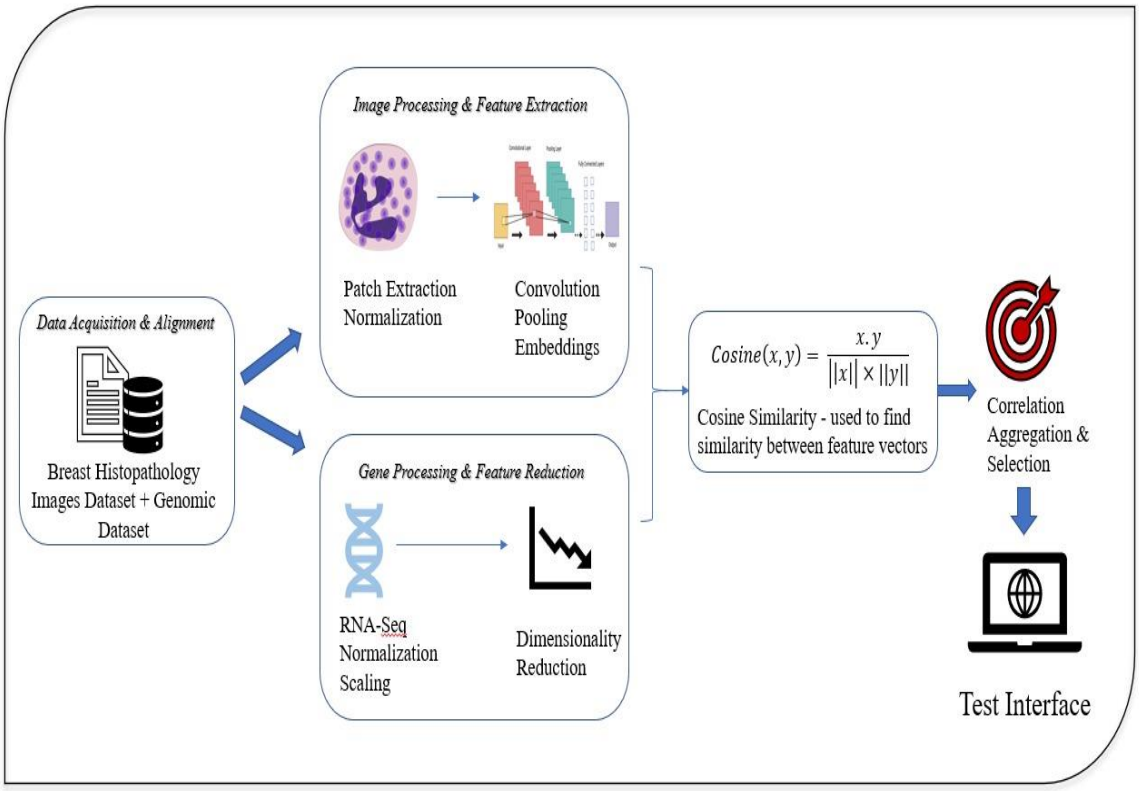


Figure 3.1: Architecture Diagram of Proposed System

3.3 Description of Algorithms

3.3.1 Support Vector Machine (SVM)

The Support Vector Machine is an useful tool for machine learning. It is used to classify things and make predictions. The main goal of the Support Vector Machine is to find a line that separates things into groups with the biggest gap possible. This gap is the distance between the line and the closest things in each group. By making this gap as big as possible the model becomes more accurate and reliable when used with information.

When we look at real world data like the information from genes it is often not possible to separate it into groups with a line. To solve this problem the Support Vector

Machine uses a trick called the kernel trick. This trick changes the data into a form that can be separated into groups with a straight line. There are a common ways to do this:

- Linear Kernel, which is used for simple data
- Polynomial Kernel, which looks at how things interact with each other
- Radial Basis Function Kernel, which is often used for data that is not in a straight line

Input: Gene expression dataset X, labels Y

Initialize $\alpha = 0$, $b = 0$

For each sample xi:

$f(xi) = \Sigma (\alpha j * yj * K(xj , xi)) + b$

If $yi * f(xi) < 1$:

$\alpha i = \alpha i + \eta$

$b = b + \eta * yi$

Output: Decision boundary separating classes

The Support Vector Machine is especially good at handling data with a lot of features like the information from genes. It can do this without getting confused by the amount of information. The Support Vector Machine also has a parameter called C, which helps find a balance between making the gap as big as possible and avoiding mistakes. This helps prevent the model from getting too complicated.

For example the Support Vector Machine can be used to tell the difference between non-cancerous samples by looking at the information from genes. It is very good at working with amounts of data and a lot of features which makes it a popular choice for biomedical research.

3.3.2 Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron is a type of neural network that can learn complex relationships between information and results. It has three parts:

- Input Layer, which receives the information

- Hidden Layers, which do calculations and change the information
- Output Layer, which gives the final result

Each part of the network is connected to the next part and each connection affects the final result. To make the relationships more complex activation functions are used at each part.

One popular activation function is the ReLU function:

$A(x) = \max(0, x)$ ———————— *equation(3.1)*

This function helps to fix some problems. Makes the training process faster. Other activation functions, like Tanh and Sigmoid can also be used depending on the problem. The Multi-Layer Perceptron is trained using a method called backpropagation, where mistakes are sent backwards through the network to update the connections. This allows the model to improve its predictions over time. In the analysis of data the Multi-Layer Perceptron is very effective because it can model complex relationships between genes. These relationships are important for understanding how cancer works. However the Multi-Layer Perceptron needs adjustment of parameters such as:

- Number of hidden layers
- Number of parts
- Learning rate
- Regularization techniques

Input: Feature vector X

Forward pass:

$z = W * X + b$

$h = \mathrm{ReLU}(z)$

Output layer:

$y_pred = \mathrm{Softmax}(W2 * h + b2)$

Backpropagation:

$W = W - \eta * \partial \mathrm{Loss}/\partial W$

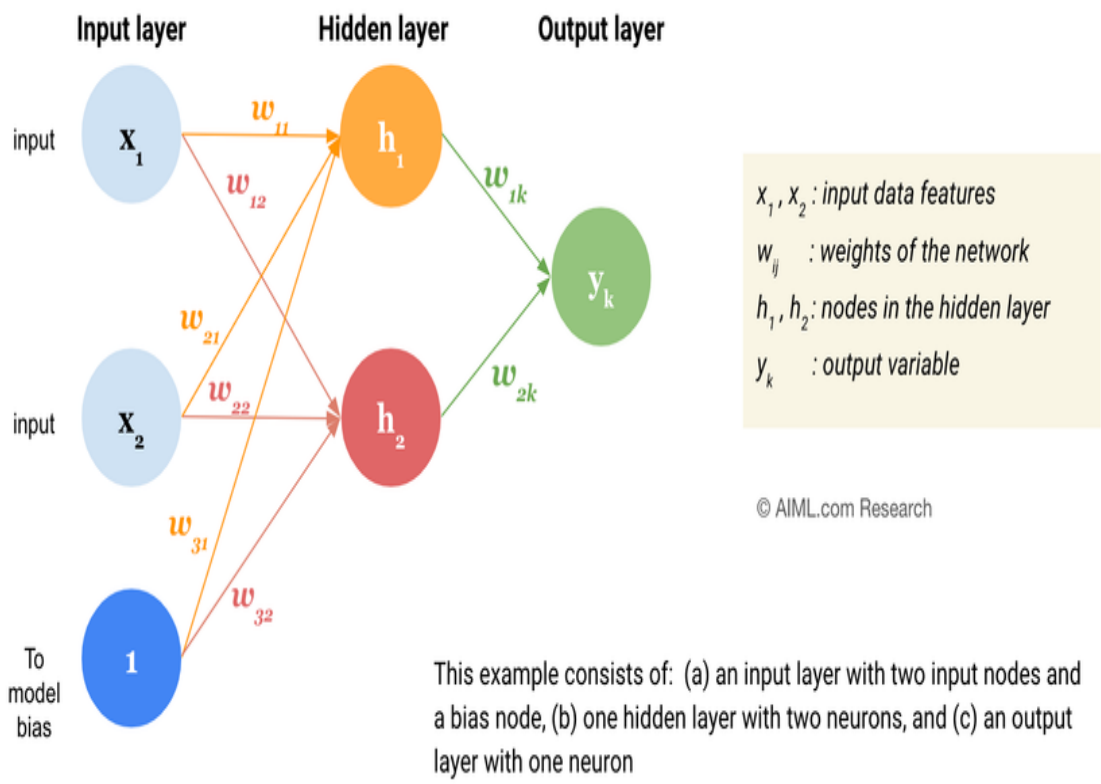


Figure 3.2: MLP Architecture

When trained properly the Multi-Layer Perceptron delivers strong results and can capture intricate biological relationships. The Multi-Layer Perceptron and the Support Vector Machine are both tools, for machine learning and can be used to analyze complex data.

3.3.3 Convolutional Neural Network (CNN)

A Convolutional Neural Network or CNN is a type of computer vision model. CNNs are great at looking at images and finding patterns. In this project we use CNNs to look at pictures of tumors and find features that can help us understand them.

Here is how a CNN works:

- Convolutional Layers: These layers look at parts of the image to find simple things like edges and textures.
- Activation Functions: We use something called ReLU to help the model learn.
- Pooling Layers: These layers help reduce the amount of data and make the model run faster.
- Connected Layers: These layers help the model make a final decision.

The CNN model is good at learning from images because it:

- Uses the filter to look at different parts of the image
- Only looks at small parts of the image at a time

Input: Tumor image I

F = Conv(I , W)

A = ReLU(F)

P = MaxPool(A)

Flatten(P) → feature vector v

Output: Feature embedding v

Although CNNs are mainly used for images we can also use them for other types of data like gene sequences [4]. In our system CNNs help us create a summary of the features of tumor images. We can then use this summary to compare with data.

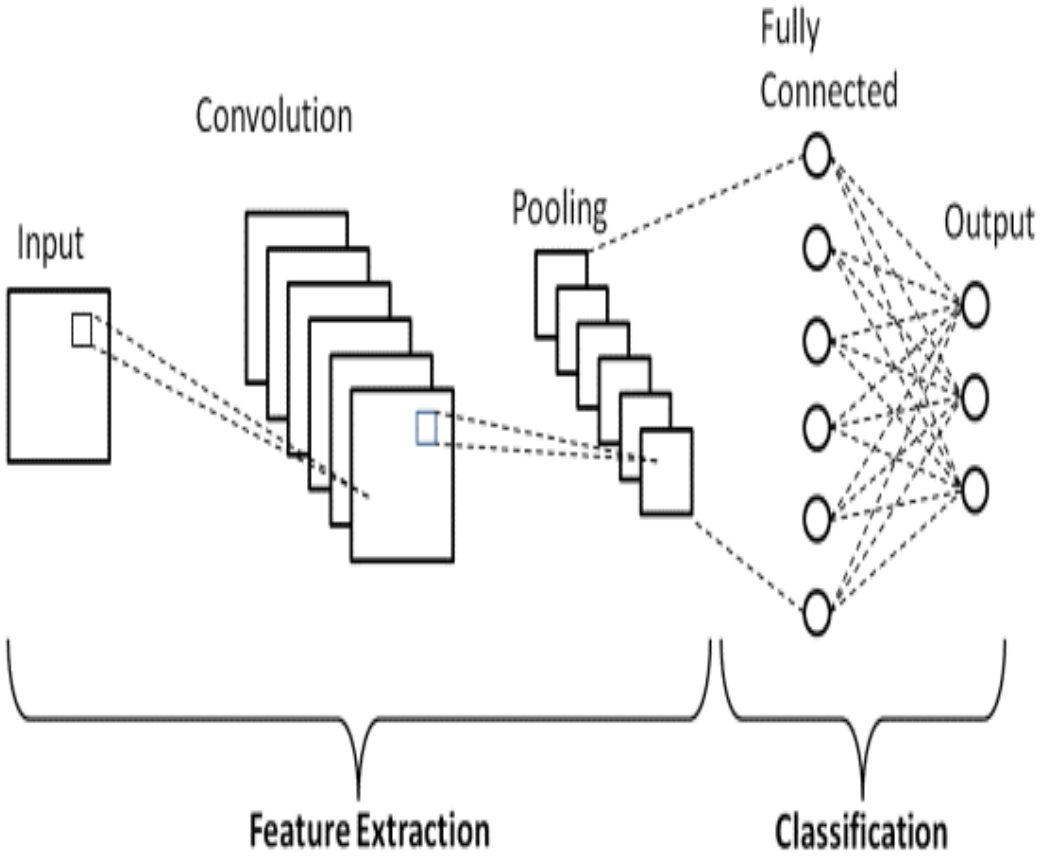


Figure 3.3: CNN Architecture

3.3.4 Deep Neural Network (DNN)

A Deep Neural Network or DNN is a type of computer model that can learn from data. DNNs are great at looking at data and finding patterns.

Here is how a DNN works:

- It looks at the data in layers starting with things and moving on to more complex things.
- The lower layers learn patterns.
- The middle layers learn combinations of features.
- The higher layers learn representations.

Input: Gene expression vector X

For each hidden layer l:

zl = Wl * al−1 + bl

al = Activation(zl)

Output:

y_pred = Softmax(aL)

DNNs are great for:

- Looking at complex datasets like gene expression profiles.
- Handling types of data.

However DNNs also need:

- A lot of data to learn from.
- A lot of computer power.
- Special techniques to prevent overfitting.

In breast cancer classification DNNs can help us find patterns, in gene expression data that can lead to prediction accuracy.

3.3.5 U-Net Algorithm

The U-Net algorithm involves the usage of a specific CNN architecture aimed at performing image segmentation. The algorithm is used in the field of biomedical image processing because of its ability to detect areas of interest. The key strength of the U-Net lies in its architecture that combines the encoder-decoder and skip connections, allowing the simultaneous acquisition of context and location information. At first, the mammogram image is fed into the encoder part. The contracting path is made of

convolutional layers with ReLU activations for feature extraction and max-pooling layers for reducing input sizes and increasing the number of feature map channels. Thus, it allows extracting hierarchical features from low-level to high-level abstractions.

The next step of the U-Net architecture is the utilization of the bottleneck layer, which helps extract high-level features from the input image and produce an encoded compact feature representation including key information needed for further processing. After that, the decoder part is responsible for producing a final segmented image. In particular, it uses upsampling to restore features. However, the crucial point of this process lies in the usage of skip connections when the feature maps from the encoder are concatenated with corresponding layers in the decoder. Thus, the spatial information is retained in contrast to being lost during down sampling in the encoder.

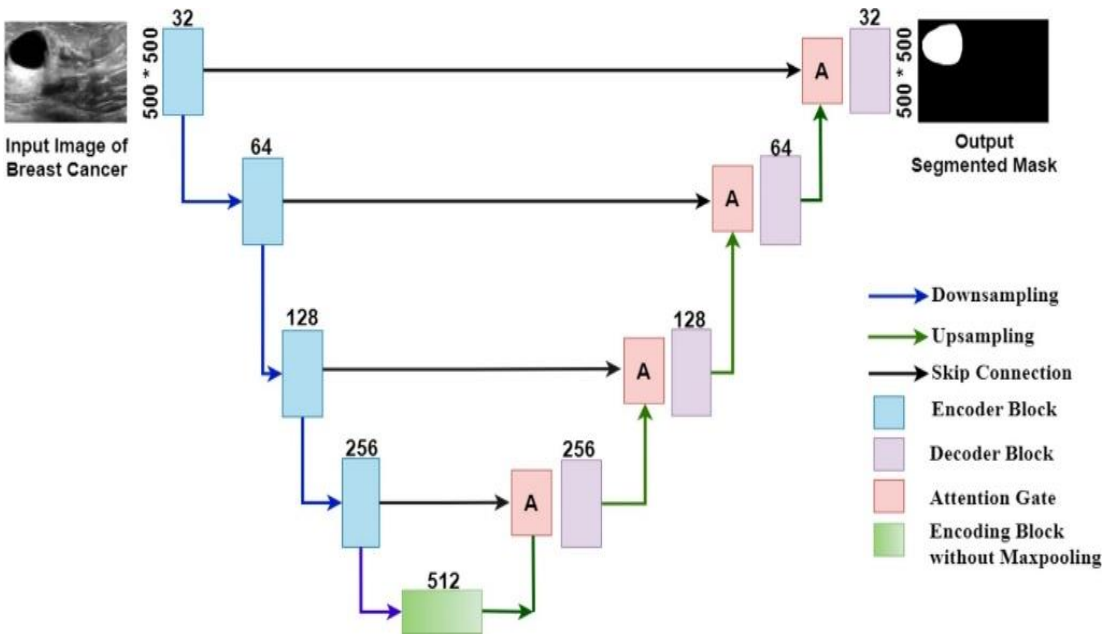


Figure 3.4: U-net Architecture

Input: Mammogram image I

Encoder:

E = Conv(I) → ReLU → MaxPool

Bottleneck:

B = Conv(E)

Decoder:

D = UpSample(B)

D = Concatenate(D , Encoder Features)

Output:

Mask M = Sigmoid(Conv(D))

Next, a 1×1 convolution maps a learned feature representation to the number of classes. Afterward, a sigmoid function produces pixel-wise probabilities of belonging to classes. Hence, the algorithm produces an output image mask identifying the presence of tumors. An output mask plays an important role in segmenting an image, making it possible to separate a region of interest from the rest of the image. Thus, it is helpful for improving the performance of the classification algorithm.

3.3.6 Relevance-Aware Capsule Network (RACapsNet)

RACapsNet is a deep learning model (i.e., Relevance-Aware Capsule Network) used to classify segmented mammographic lesions as either benign or malignant. It is a modified version of a traditional capsule network (CapsNet) that was designed to improve the accuracy and interpretability of medical diagnosis using deep learning. Capsule networks use vector representations (or capsules), to accurately represent the spatial relationships between features, as opposed to traditional convolutional neural networks (CNNs), which use only scalar representations for their outputs. RACapsNet extends the capsule idea further by adding the Relevance layer to aid in the explanation of the decisions made by the model through highlighting the relevant areas within the input image.

Classification begins with a tumor area that has already been segmented through the U-Net model. This ensures that the classification is of only the tumor area, and not the surrounding areas or other structures. Once the tumor area is established, the input is passed through a series of convolutional layers to extract low-level features and generate primary capsules containing the presence and spatial information (i.e., orientation and position) of the extracted features.

Input: ROI feature map F

Primary capsules:

U = Caps(F)

Prediction vectors:

u_hat = W * U

Dynamic routing:

v = squash(Σ(cij * u_hat))

Output:

y_pred = argmax ||v||

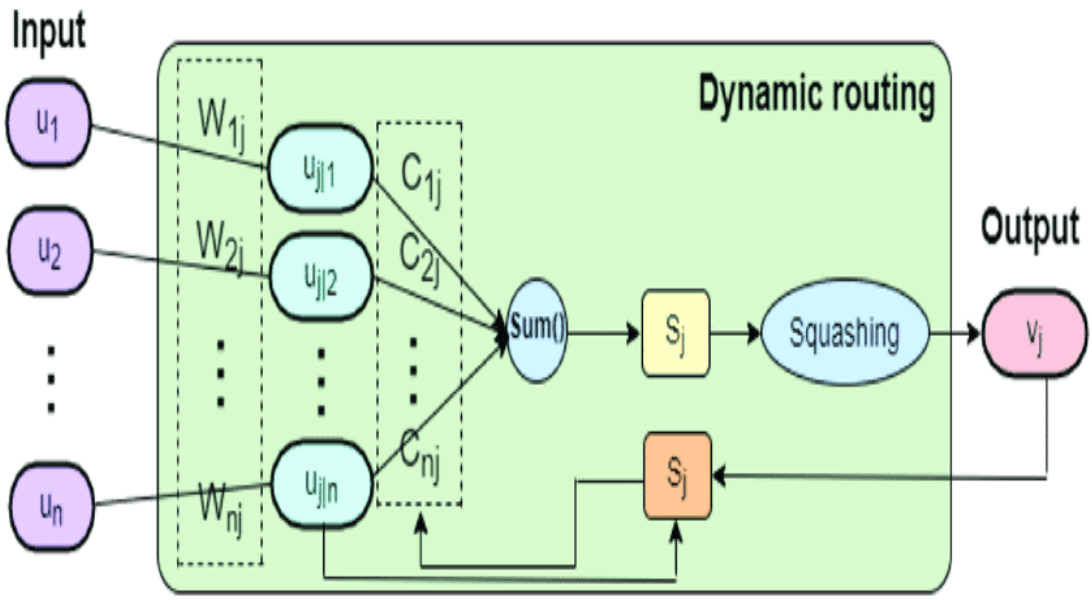


Figure 3.5: Routing in Capsule Network

RACapsNet also contains a dynamic routing architecture, which is used to determine the transfer of information between low and high-level capsules. Dynamic routing is used to adjust the routing coefficients in the model based on the agreement of the adjacent capsules as opposed to a static routing structure used in traditional networks. This allows the network to capture complex spatial hierarchies and relationships within the image, making it more robust than traditional models.

Capsule Squashing Function (RACapsnet):

$$v_j = \frac{||s_j||^2}{1+||s_j||^2} \cdot \frac{s_j}{||s_j||} \qquad \text{equation(3.2)}$$

Where : s_j = capsule input, v_j = output vector

Dynamic Routing Agreement:

$$b_{ij} \leftarrow b_{ij} + \hat{u}_{j|i} \cdot v_j \qquad \text{———————} \quad \textit{equation(3.3)}$$

Where:

b_{ij} = routing coefficient between capsule i and j

$u_{j|i} = W_{ij\ Ui}$ = prediction vector

v_j = output of higher capsule

3.3.7 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is one of the statistical methods used to reduce dimensions of a dataset. In other words, PCA allows converting a high-dimensional dataset to a small set of variables called principal components that preserve the maximal variance of data. Thus, the goals of PCA include:

- Reducing the number of features
- Removing redundancy and noise
- Preserving important information

Input: Feature matrix X

Compute covariance:

C = cov(X)

Eigen decomposition:

C * v = λ * v

Select top k eigenvectors:

W = [v1 , v2 , ... , vk]

Reduced data:

Z = X * W

Specifically, the application of PCA within this research includes:

a. Genomic Data Reduction

In many cases, gene expression datasets have thousands of genes, the majority of which can be redundant. Therefore, the use of PCA allows obtaining a low-dimensional representation of the data, while preserving its biological meaning.

b. Image Feature Reduction

Feature vectors produced by Convolutional Neural Networks (CNNs) can be large. To address this issue, PCA is used in order to:

- Remove redundant features
- Make calculations more efficient
- Improve similarity measures

With the help of PCA, it becomes possible to achieve:

- Avoidance of overfitting
- Performance improvement
- Interpretation enhancement

3.3.8 Cosine Similarity

Cosine similarity is a mathematical concept that can be used for calculating the similarity of two vectors depending on the angle between them. As distinct from the distance-based measures, cosine similarity considers direction of vectors rather than magnitude, making it relevant for analyzing high-dimensional datasets. The values of cosine similarity range from:

- +1 – highly similar (high correlation)
- 0 – not similar
- -1 – opposite directions

$$Cosine(x, y) = \frac{x.y}{||x|| \times ||y||} \quad \text{———————} \quad \textit{equation (3.4)}$$

Input: Image feature vector X, Genomic feature vector Y

dot = Σ (Xi * Yi)

norm_x = sqrt(Σ Xi^2)

norm_y = sqrt(Σ Yi^2)

cos_sim = dot / (norm_x * norm_y)

Output: Similarity score

Thus, cosine similarity will play an important role in multimodal analysis of this project. Specifically, cosine similarity will be used for comparing genomic feature vectors (resulting from PCA) and image feature vectors (obtained using CNN). This method will allow finding similarities when some gene expression pattern corresponds to certain tissue structure.

3.4 Description of datasets, Requirement specification

3.4.1 Datasets

Breast Histopathological Dataset refers to images at high resolution of biopsy-sampled breast tissues used for detecting cancer based on studying cellular structures and tissues' morphology. Typically, such images are split into patches and labeled as either benign (non-cancerous) or malignant (cancerous), with different sizes, shapes, and staining intensities represented at several magnification levels, including 40x, 100x, 200x, and 400x. Preprocessing for these images may include tasks such as resizing, normalization, augmentation (including rotation, flipping, scaling), and segmentation. The dataset is popular for training models to detect, classify, and extract features in tumors by means of deep learning architectures, such as CNNs and U-Net.

On the other hand, the TCGA-BRCA dataset, collected within the project The Cancer Genome Atlas, includes genomic data of breast cancer patients in the form of RNA-Seq gene expression profiles, containing information about both tumor and normal tissue types together with additional clinical metadata (patient ID, tumor stage, and other characteristics) [3]. The dataset is characterized by extreme high dimensionality, with numerous gene expression values per sample and significant levels of missing or noisy data. As a result, additional preprocessing tasks such as normalization, missing value imputation (for example, with GAIN algorithm), feature selection, and dimensionality reduction (such as PCA) are often required. Applications of this dataset include cancer classification, biomarkers discovery, and gene expression analysis.

While in the first case we get structural tissue features in the form of images, in the second case the information we obtain is at the molecular level – namely, the gene expression profile of a patient. Combining them, we get a multimodal approach for obtaining a better insight into the structure of breast cancer by analyzing both phenotypic and genotypic aspects of it. When performing a correlation-aware multimodal analysis of the two datasets, they can serve as complementary input sources. The first one provides information related to the visual aspect – cell arrangement, tissue abnormalities, and micro-environment structure. The second source provides the information about molecular mechanisms behind tumor development. With the help of appropriate preprocessing techniques, including visual

embedding extraction from the images and dimensionality reduction of RNA-Seq data, the projections of the two data sources to a latent common space allow for quantitative assessment of the correlations between them.

3.4.2 Requirement Specification

The design and evaluation of the suggested correlation-aware multimodal analysis framework will require a suite of software tools, machine learning packages, and sufficient computing power that could handle the processing of multidimensional genomic and imaging data. The use of the Python programming language is motivated by its rich set of packages and tools for scientific computing, bioinformatics data processing, and deep learning methods. For instance, Pandas and NumPy are used to work with patient-level metadata and conduct complex computations with genomic matrices, respectively. In turn, OpenCV and SciPy are applied to preprocess images (resize, normalization, and tensor transformations).

To develop models, one will need to install deep learning packages (e.g., TensorFlow/Keras, PyTorch) to build the Convolutional Neural Network (CNN) to obtain visual embeddings from histological images. As the next step, Scikit-learn is leveraged to conduct PCA to reduce the dimensionality of vast RNA-Seq data and create informative latent representation. Moreover, cosine similarity needs to be computed as an alignment metric using additional computational logic. Visualization libraries like Matplotlib and Seaborn could help in exploring correlations, obtaining top associations between different patient samples, and showing histological images of the tumor tissue.

Since deep learning inference is resource-intensive, especially for processing hundreds of thousands of image patches, the use of GPU is necessary to accelerate the computations. It is reasonable to choose a machine with at least 16 GB of RAM and NVIDIA graphics card (8 GB VRAM). Concerning storage needs, it should be emphasized that more than 100 GB of fast SSD disk space will be needed to store both the TCGA-BRCA genomics data and many image patches of digitized tumor sections along with intermediate model weights and latent variables.

CHAPTER-4

Results and Observations

4.1 Stepwise description of Results

Original dataset (Gene Expression Omnibus) is problematic due to missing data (Figure 4.1), particularly in gene expression values across 62,976 probes. An initial sample of the first 10 rows of data provided numerous missing (NaN) values which could provide difficulty for model performance leading to inaccurate outcomes.

```
First 10 rows of the dataset:
        1      2      3      4      5      6       7       8      9     10  \
0     NaN    NaN    NaN    NaN    NaN    NaN     NaN     NaN    NaN    NaN
1     NaN    NaN    NaN  9.387  7.125  7.033  10.435   9.536  5.366  5.565
2  15.684  5.196  5.167  8.958  7.993  6.359  10.829   9.408  5.403  5.941
3  15.283  4.458  3.173  9.035  7.856  6.261  11.649   8.793  4.631  6.746
4  15.535    NaN    NaN    NaN    NaN    NaN     NaN     NaN    NaN    NaN
5  15.106  4.127  4.178  9.188  8.042  6.388  11.332   8.884  5.118  6.643
6  15.321  5.629  4.665  9.326  7.653  6.990  11.581   9.439  5.797  6.653
7     NaN    NaN    NaN    NaN    NaN    NaN     NaN     NaN    NaN    NaN
8  15.761  4.150  4.209  9.298  7.356  7.040  12.247   9.543  4.597  6.564
9  16.828  3.676  2.936  8.042  7.259  6.470  11.281  10.439  5.129  6.286

   ...  62969  62970  62971  62972  62973  62974   62975   62976      Sample  \
0  ...    NaN    NaN    NaN    NaN    NaN    NaN     NaN     NaN  GSM1823702
1  ...    NaN    NaN    NaN    NaN    NaN    NaN     NaN     NaN  GSM1823703
2  ...    NaN    NaN    NaN    NaN    NaN    NaN     NaN     NaN  GSM1823704
3  ...  9.044  5.098  7.424  6.216  4.631  4.024  15.469  15.447  GSM1823705
4  ...  8.376    NaN    NaN    NaN  4.496  4.295  15.509  15.491  GSM1823706
5  ...    NaN    NaN    NaN    NaN    NaN    NaN     NaN     NaN  GSM1823707
6  ...  8.551  4.818  6.202  6.310  4.508  4.818  15.272  15.387  GSM1823708
7  ...  8.817  5.330  6.814  5.740  4.455  4.560  15.261  15.207  GSM1823709
8  ...    NaN    NaN    NaN    NaN    NaN    NaN     NaN     NaN  GSM1823710
9  ...  8.815  5.310  7.178  6.588  5.129  4.329  16.393  16.858  GSM1823711

              Condition
0  sample id: CM016.N.c2
1  sample id: CM022.N.c2
2  sample id: CM082.N.c2
3  sample id: CM093.N.c2
4  sample id: CM096.N.c2
5  sample id: CM099.N.c2
6  sample id: CM101.N.c2
7  sample id: CM103.N.c2
8  sample id: CM106.N.c2
9  sample id: CM113.N.c2
```

Figure 4.1: Before Handling Missing Values

In order to address the missing data issue, we used an advanced method of imputation called GAIN (Generative Adversarial Imputation Nets) , which uses deep learning methods to make imputation predictions based on learned patterns from the remaining data [9]. GAIN not only maintains the statistical properties of the original

data but also provides more realistic estimates of the missing data. Visualizing data post-processing (Figure 4.2) provides evidence that the imputed data is now complete (no missing data); and that all originally missing values have been calculated based upon the patterns of the existing data.

The expense of working with 62,976 gene expression features places a strong computational burden on the model, and increases the likelihood of overfitting. Thus, several dimensionality reduction techniques (PCA), (Variance Thresholding), and (Select K-Best) were employed to retain only the most informative features.

The final dataset consisted of 18 gene features across an equal distribution of the outcome class labels: 148 cancer (1); and 148 non-cancer (0) samples.

```
First 10 rows after handling missing values:
          1         2         3         4         5         6          7  \
0  14.764448  4.178353  4.182095  8.647605  7.314619  6.395718  11.313014
1  14.764448  4.178353  4.182095  9.387000  7.125000  7.033000  10.435000
2  15.684000  5.196000  5.167000  8.958000  7.993000  6.359000  10.829000
3  15.283000  4.458000  3.173000  9.035000  7.856000  6.261000  11.649000
4  15.535000  4.178353  4.182095  8.647605  7.314619  6.395718  11.313014
5  15.106000  4.127000  4.178000  9.188000  8.042000  6.388000  11.332000
6  15.321000  5.629000  4.665000  9.326000  7.653000  6.990000  11.581000
7  14.764448  4.178353  4.182095  8.647605  7.314619  6.395718  11.313014
8  15.761000  4.150000  4.209000  9.298000  7.356000  7.040000  12.247000
9  16.828000  3.676000  2.936000  8.042000  7.259000  6.470000  11.281000

          8         9        10  ...     62969     62970     62971     62972  \
0   9.142224  4.903296  6.243218  ...  8.014059  4.754959  5.866129  5.489826
1   9.536000  5.366000  5.565000  ...  8.014059  4.754959  5.866129  5.489826
2   9.408000  5.403000  5.941000  ...  8.014059  4.754959  5.866129  5.489826
3   8.793000  4.631000  6.746000  ...  9.044000  5.098000  7.424000  6.216000
4   9.142224  4.903296  6.243218  ...  8.376000  4.754959  5.866129  5.489826
5   8.884000  5.118000  6.643000  ...  8.014059  4.754959  5.866129  5.489826
6   9.439000  5.797000  6.653000  ...  8.551000  4.818000  6.202000  6.310000
7   9.142224  4.903296  6.243218  ...  8.817000  5.330000  6.814000  5.740000
8   9.543000  4.597000  6.564000  ...  8.014059  4.754959  5.866129  5.489826
9  10.439000  5.129000  6.286000  ...  8.815000  5.310000  7.178000  6.588000

      62973    62974      62975      62976      Sample              Condition
0  3.734101  3.81271  14.291158  14.169943  GSM1823702  sample id: CM016.N.c2
1  3.734101  3.81271  14.291158  14.169943  GSM1823703  sample id: CM022.N.c2
2  3.734101  3.81271  14.291158  14.169943  GSM1823704  sample id: CM082.N.c2
3  4.631000  4.02400  15.469000  15.447000  GSM1823705  sample id: CM093.N.c2
4  4.496000  4.29500  15.509000  15.491000  GSM1823706  sample id: CM096.N.c2
5  3.734101  3.81271  14.291158  14.169943  GSM1823707  sample id: CM099.N.c2
6  4.508000  4.81800  15.272000  15.387000  GSM1823708  sample id: CM101.N.c2
7  4.455000  4.56000  15.261000  15.207000  GSM1823709  sample id: CM103.N.c2
8  3.734101  3.81271  14.291158  14.169943  GSM1823710  sample id: CM106.N.c2
9  5.129000  4.32900  16.393000  16.858000  GSM1823711  sample id: CM113.N.c2
```

Figure 4.2: After Handling Missing Values Using GAIN

The results of the class distribution bar graph illustrated in figure 4.3 provided evidence that perfect class balance was achieved in the dataset. As such, the classification model would not suffer from class bias.

```
Final reduced shape: (296, 53)
```

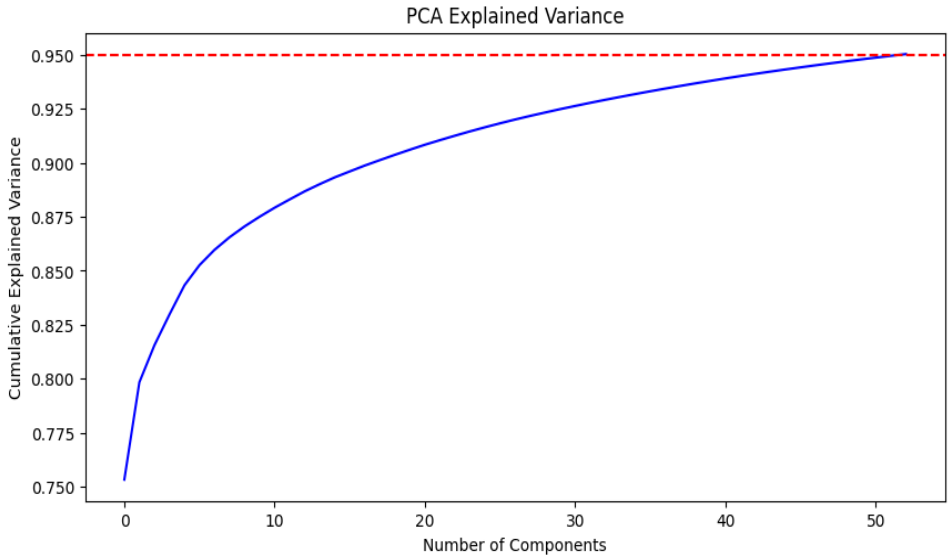


Figure 4.3 Total features after applying feature reduction

CTGAN was then used to generate synthetic gene expression data, as shown in figure 4.4, to enhance both the robustness and generalisability of the model, through the augmentation of the dataset by creating:

- 352 synthetic cancer samples
- 352 synthetic normal samples

```
Class labels after feature reduction:
Total samples: 296
Cancer samples (1): 148
Non-cancer samples (0): 148
```

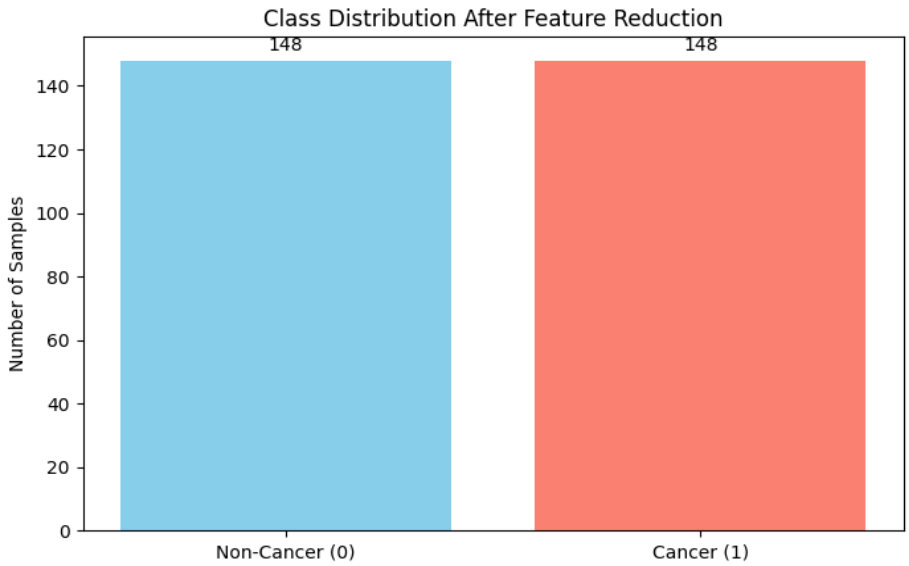


Figure 4.4 Raw Dataset Size

Thus, creating a dataset with 1,000 total samples (500 in each class). The CTGAN was trained independently for the cancer and non-cancer classes and used adversarial learning; whereby both generator and discriminator networks improved their respective data quality through iterative training (figure 4.5). Logs from CTGAN showed convergence and successful training with discriminator scores approaching 100%, indicating that high quality synthetic samples were being produced by the generator.

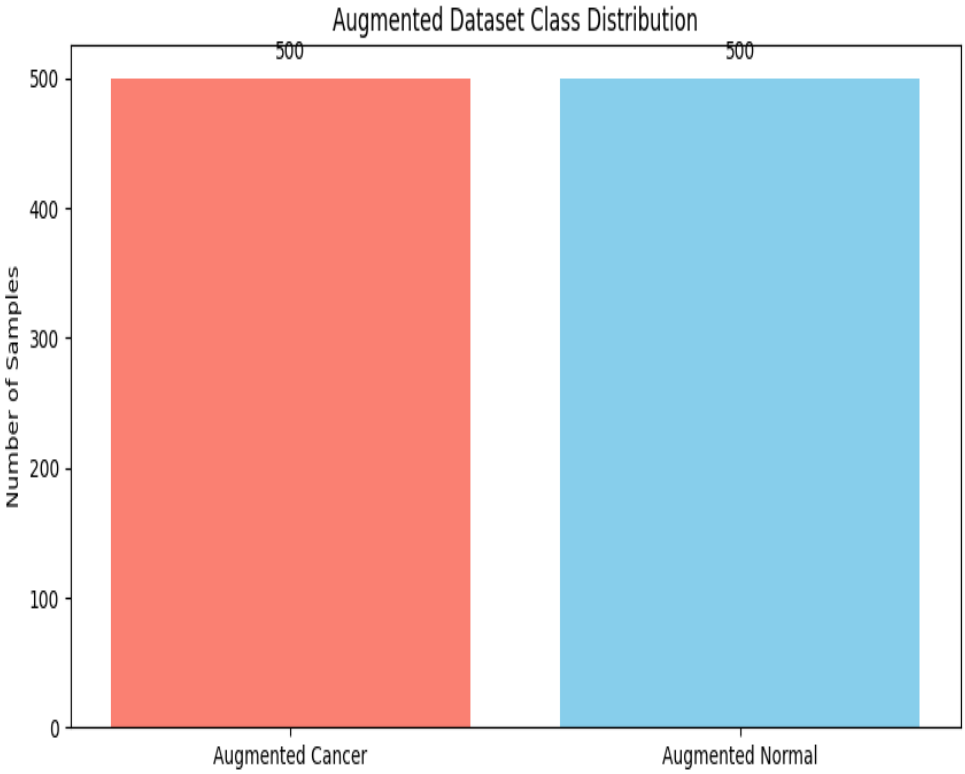


Figure 4.5: Augmented Dataset Using CTGAN

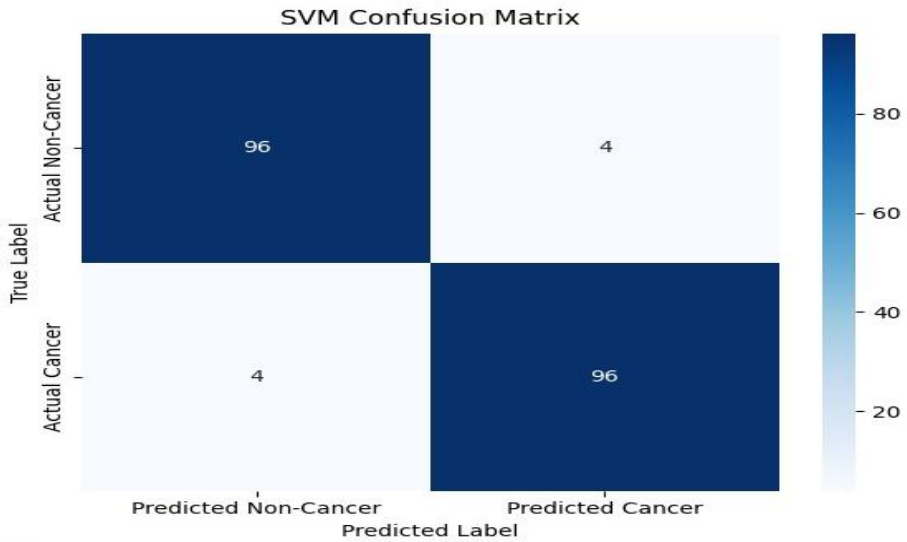
Once data augmentation had been completed the balanced data was reviewed using an SVM classifier that employed the RBF kernel to complete this task, as shown in Figure 4.6 below. The model was assessed using several different metrics: Precision: .96 Recall: .96 F1-Score: .96 Accuracy: 96%

By examining the results of the confusion matrix, we find that this model produced 192 correct classifications and 8 incorrect classifications. Of the 200 testing instances, there were 4 false positive classifications and 4 false negative classifications. This level of accuracy indicates that our model was able to generalize to both classes due to the pre-processing techniques and augmentation techniques that were used.

```
Classification Report:
              precision    recall  f1-score   support

  Non-Cancer       0.96      0.96      0.96       100
      Cancer       0.96      0.96      0.96       100

    accuracy                           0.96       200
   macro avg       0.96      0.96      0.96       200
weighted avg       0.96      0.96      0.96       200
```




```
Final SVM Accuracy: 96.00%
```

Figure 4.6: Results for Classification using Support Vector Machine (SVM)

The MLP model's performance can be reviewed in great detail using accuracy measures; therefore, the MLP model's performance is exceptional as it achieved an accuracy of 99.0% for training, 98.75% for validation, and 98.0% for testing through 33 epochs of training. These accuracies illustrate that the MLP model has a solid learning capability on the data used during training and an excellent generalization capability for data not previously seen.

The classification report further supports this conclusion. The performance metrics achieved by this model are :

Accuracy: $\frac{1}{|Te|}\sum_{x\epsilon Te} I[\,\hat{C}(x) = c(x)] = 99\%$ ——————— *equation(4.1)*

Precision: $\frac{\sum_{x\epsilon Te} I[\hat{C}(x)=C(x)=+]}{\sum_{x\epsilon Te} I[\hat{C}(x)=+]} = 98\%$ ——————— *equation(4.2)*

Recall: $\frac{\sum_{x\epsilon Te} I[\hat{C}(x)=C(x)=+]}{\sum_{x\epsilon Te} I[C(x)=+]} = 98\%$ ——————— *equation(4.3)*

F1-Score: $2 \times \frac{precision*recall}{precision+recall} = 98\%$ ——————— *equation(4.4)*

```
🎯 Accuracy Summary:
Training Accuracy   : 0.9900
Validation Accuracy : 0.9875
Test Accuracy       : 0.9800
Epochs Trained      : 33

Classification Report:
              precision    recall  f1-score   support

           0       0.97      0.99      0.98       100
           1       0.99      0.97      0.98       100

    accuracy                           0.98       200
   macro avg       0.98      0.98      0.98       200
weighted avg       0.98      0.98      0.98       200
```

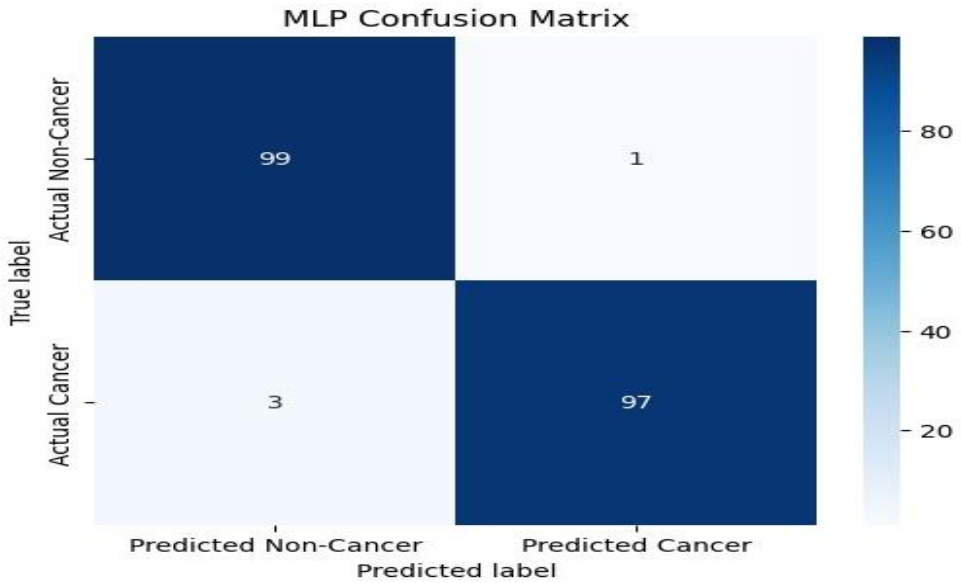


Figure 4.7: Results for Classification using Multi-Layer Perceptron (MLP)

The confusion matrix represented in Figure 4.7 gives an additional understanding of the model's ability to classify. Out of 100 Non-Cancer cases, 99 were classified right while one of them was misclassified. Out of 100 Cancer cases, 97 of them were classified right (as Cancer) while three of them were misclassified (as non-Cancer). This means there were 97 True Positives, 3 False Negatives, 99 True Negatives, and 1 False Positive. Thus, there were 4 misclassifications out of 200 classifications which implies the high accuracy and reliability of the model, and hence its applicability in crucial areas like cancer detection.

Figure 4.8 shows how an increase in the number of training examples influences the model's accuracy.

- X-axis: Number of training examples (between approximately 60 and 600).
- Y-axis: Accuracy score (between approximately 0.60 and 1.00).

- Blue line (Training): Throughout the whole training process, accuracy was always high (95%-99%), which suggests a good fit between the model and training data.
- Red line (Validation): With more training examples, validation accuracy increased and reached around 95%.

To sum up, the model showed excellent results with the growing number of examples without significant overfitting. Despite its somewhat worse results than those shown by SVM, it still managed to detect non-linearity within the dataset.

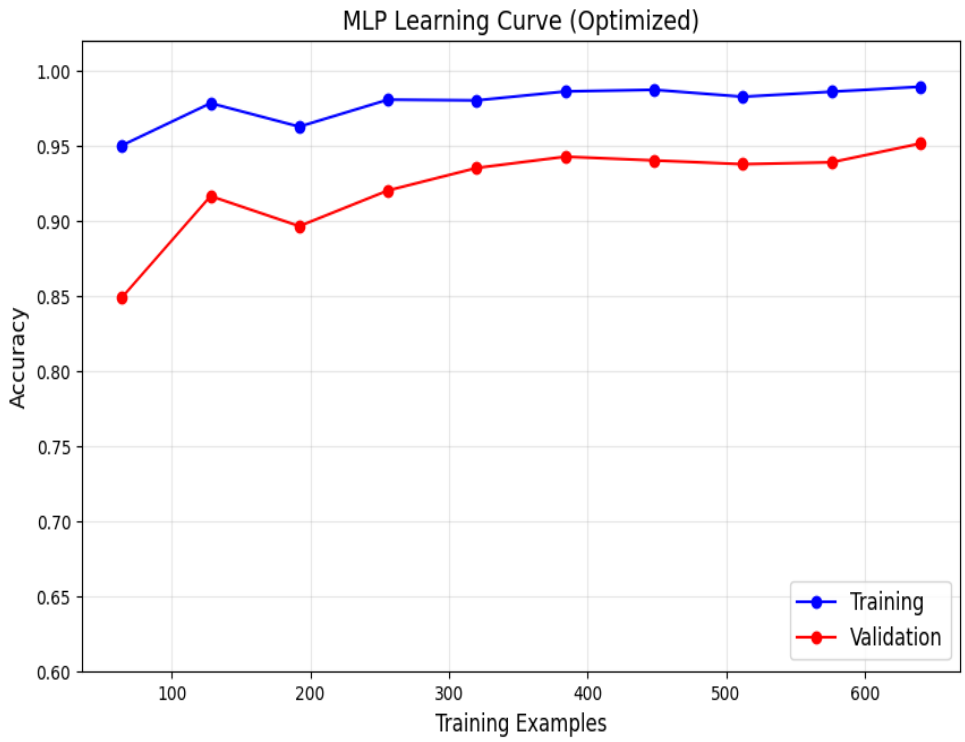


Figure 4.8: MLP Learning Curve

The results provided by the model of CNN (Figure 4.9) show great classification capabilities with the overall accuracy equal to 91%. Classification report presents detailed classification information regarding two classes: Non-Cancer and Cancer. Regarding the Non-Cancer class, the algorithm showed precision value equal to 0.88, recall equal to 0.95 and F1-score equal to 0.91, which means that the majority of cases have been detected with relatively few false positives. As far as the Cancer class is concerned, precision value is higher (0.95) and recall value is lower (0.87) compared to the other class, resulting in F1-score of 0.91. Thus, it can be stated that the model is

extremely efficient in predicting cancer cases correctly (precision); however, there are instances of missed cancer cases (recall). It should be noted that the averages of all metrics for macron and weighted metrics are the same, namely 0.91.

The information presented by the confusion matrix shows the model's predictions more clearly. Out of 100 actual Non-Cancer cases, the model has predicted 95 of them correctly and has missed 5 cases and misclassified them as Cancer. Out of 100 actual Cancer cases, the model has predicted 87 of them correctly and has misclassified 13 cases as Non-Cancer. Based on the results presented above, it follows that there are 87 true positives (Cancer detected), 13 false negatives (Cancer missed), 95 true negatives (Non-Cancer detected), and 5 false positives (Non-Cancer missdetected). Taking into account the high number of missed cancer cases, it should be mentioned that it is important to improve the model's performance since missing cancer cases can cause serious issues. Overall, the model operates rather efficiently, but it needs some improvements related to the issue of recall.

Classification Report (CNN):

	precision	recall	f1-score	support
Non-Cancer	0.88	0.95	0.91	100
Cancer	0.95	0.87	0.91	100
accuracy			0.91	200
macro avg	0.91	0.91	0.91	200
weighted avg	0.91	0.91	0.91	200

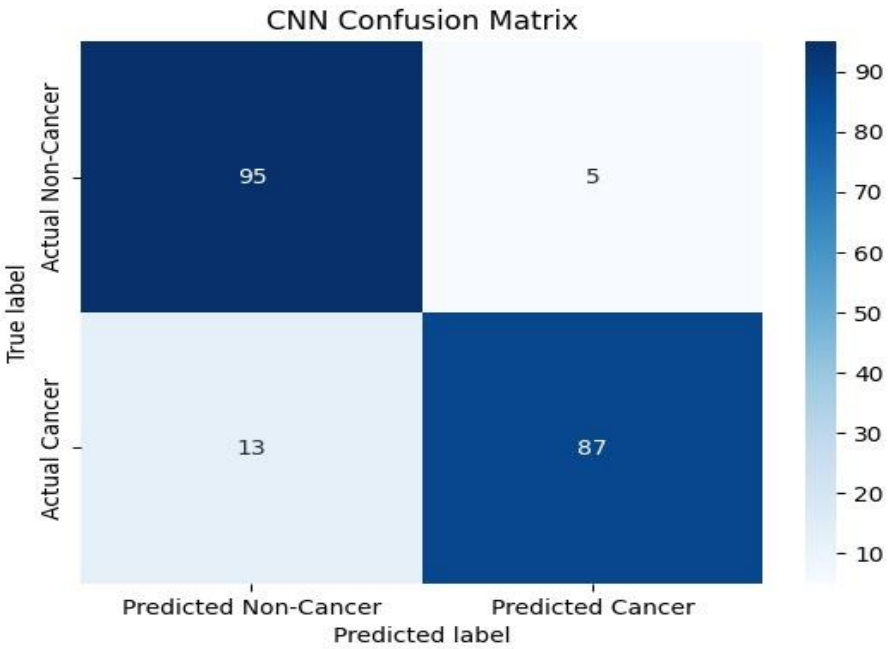


Figure 4.9: Results for Classification using Convolution Neural Network (CNN)

As far as the learning curve (Figure 4.10) is concerned, the following conclusions can be made based on it:

- There is a significant improvement of training accuracy that tends towards 95%;
- Validation accuracy starts from a low level, decreases first (it is typical for the beginning of training process), then rises up to 88%-89%;
- No problems with overfitting.

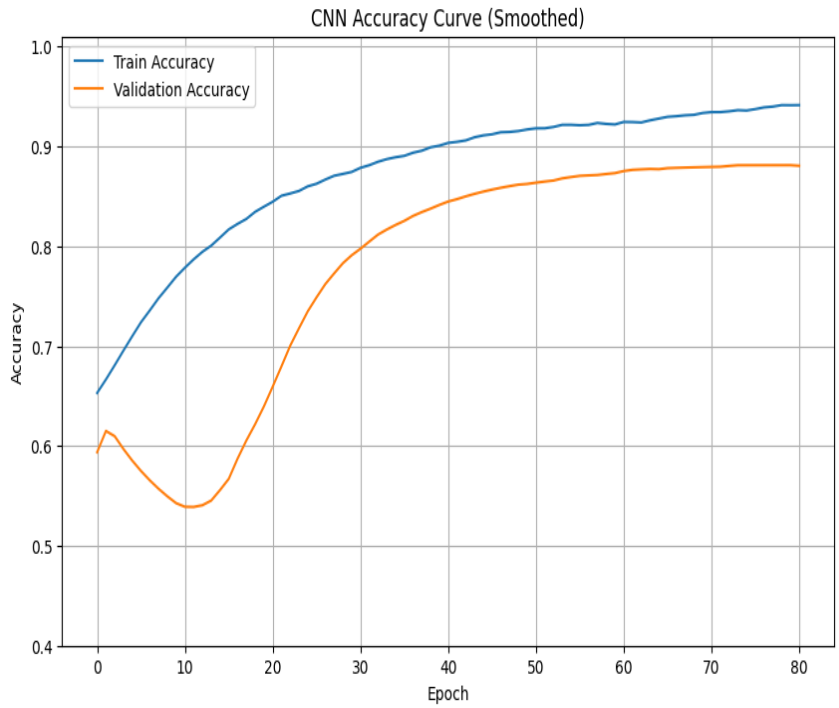


Figure 4.10: CNN Accuracy Curve

First, the classification report generated for the DNN model indicates high performance since its overall accuracy reaches 96% (refer to Fig. 4.11 below). Thus, in terms of the Non-Cancer class (or Class 0), the classifier's precision is equal to 0.94, while recall equals 0.99, and F1-score amounts to 0.97, which means that it demonstrates extremely high effectiveness for the identification of non-cancerous objects and almost no false-negative results. As far as the Cancer class (or Class 1) is concerned, then precision amounts to 0.99; moreover, recall reaches 0.94, whereas F1-score equals 0.96, and that means that the DNN classifier is very efficient when it comes to minimizing false-positive results, which is critical when it comes to applying AI algorithms for diagnostic purposes. In turn, the macro and weighted averages of precision, recall, and F1-score fall between 0.96 and 0.97.

```
Classification Report (DNN):
              precision    recall  f1-score   support

  Non-Cancer       0.94      0.99      0.97       100
      Cancer       0.99      0.94      0.96       100

    accuracy                           0.96       200
   macro avg       0.97      0.96      0.96       200
weighted avg       0.97      0.96      0.96       200
```

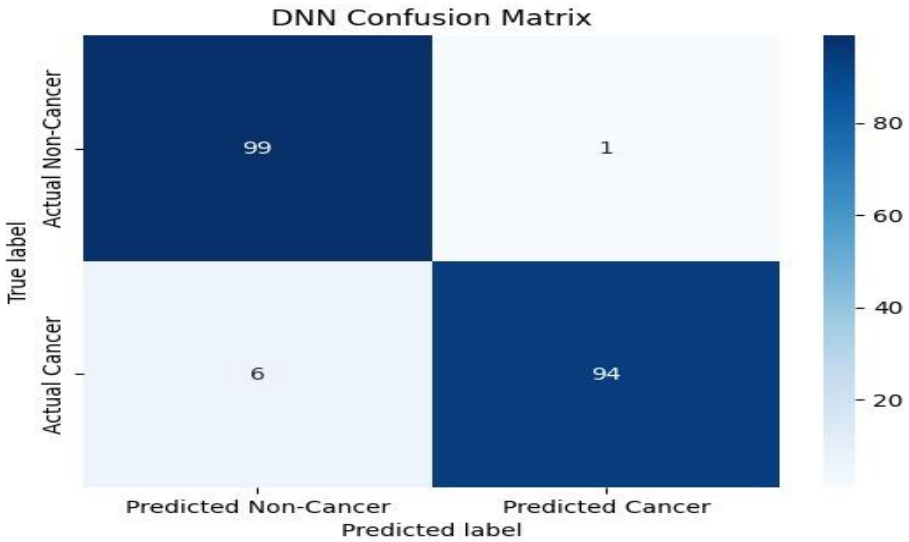


Figure 4.11: Results for Classification using Deep Neural Network (DNN)

The confusion matrix (Fig. 4.11) generated by the DNN classifier demonstrates that the classifier managed to identify all samples properly except for 7 of them out of 200 analyzed. In other words, there are 94 true positives, 99 true negatives, 6 false negatives (or wrongly classified cancers), and only one false positive (a cancer case that was mistakenly classified).

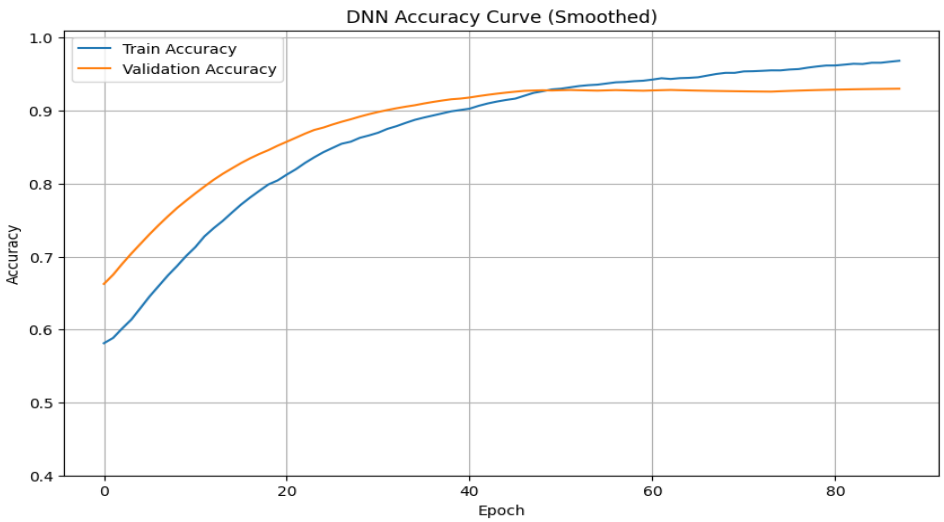


Figure 4.12: DNN Accuracy Curve

Training and validation accuracy of the classifier (Fig. 4.12) were calculated during more than 85 epochs:

- Training accuracy consistently grows to 98%
- Validation accuracy grows too and stabilizes at 93%.

The outputs of the suggested detection system are based on a certain procedure of data handling. Initially, the dataset of mammograms obtained from CBIS-DDSM was investigated. According to Figure 4.13, there is a significant class imbalance between the two classes as 1010 benign samples and 708 malignant samples exist in the CBIS-DDSM mammograms dataset. Hence, 302 additional malignant samples have to be generated to have equal numbers. The result is evident by stating "Need to generate: 302 extra malignant images". To achieve this goal, common augmentation methods including rotation, flipping, contrast, intensity manipulation, and CLAHE were used specifically for the malignant class. The resulting augmentation provided 302 extra samples that equalized the number of samples for both classes, providing 1010 benign and 1010 malignant images. Consequently, the perfectly balanced dataset was achieved by obtaining "After balancing: 1010 malignant, 1010 benign," and storing the dataset with 2020 total samples. Balancing of the dataset was critical in terms of achieving higher accuracy of the classification task.

```
Before balancing:
Benign: 1010  Malignant: 708
Need to generate: 302 extra malignant images
```

```
After balancing:
label
0    1010
1    1010
Name: count, dtype: int64
Saved train_balanced.csv with shape: (2020, 59)
```

Figure 4.13: Before and after balancing the dataset

The raw dataset obtained from the CBIS-DDSM collection contains a variety of dense, fatty, and abnormal regions with high visual complexity. In many cases, malignant areas appeared faint and partially overlapped adjacent structures. Thus, conventional augmentation approaches were used as a method to enrich the dataset with new samples and solve a problem of class imbalance. The malignant samples were augmented through rotation, flipping, contrast adjustment, intensity manipulation, and CLAHE. Figure 4.14 depicts how augmentation provides a certain level of variability

in tissue texture, lighting, and orientation while ensuring equal sample distribution for each class after balancing.

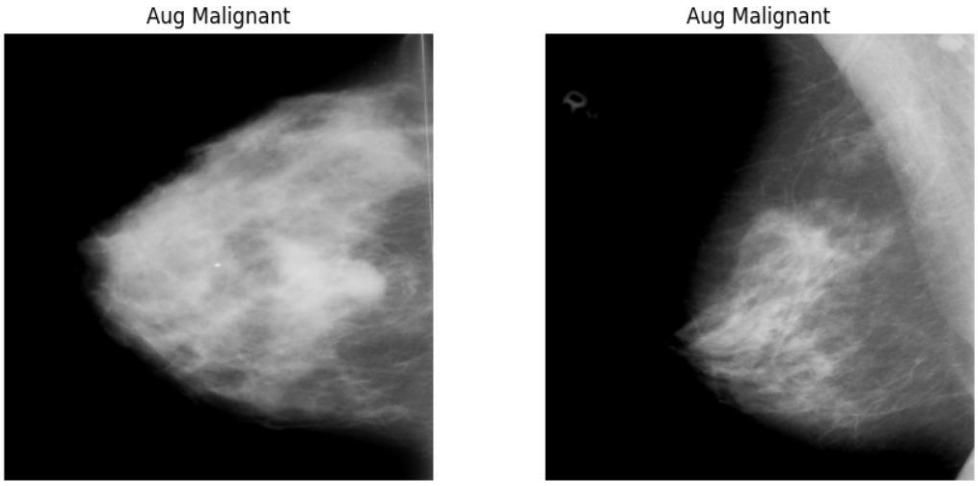


Figure 4.14: Representative Augmented Malignant Mammogram Samples

Region of interest (ROI) segmentation was carried out using a U-Net model. It could be observed from "Before ROI Segmentation" pictures that they include full mammograms representing the whole breast structure with all its complex details. With the help of a U-Net approach, the images labeled "After ROI Segmentation" show only the tumor area as a distinct white patch with the rest of the image removed (see Figure 4.15).

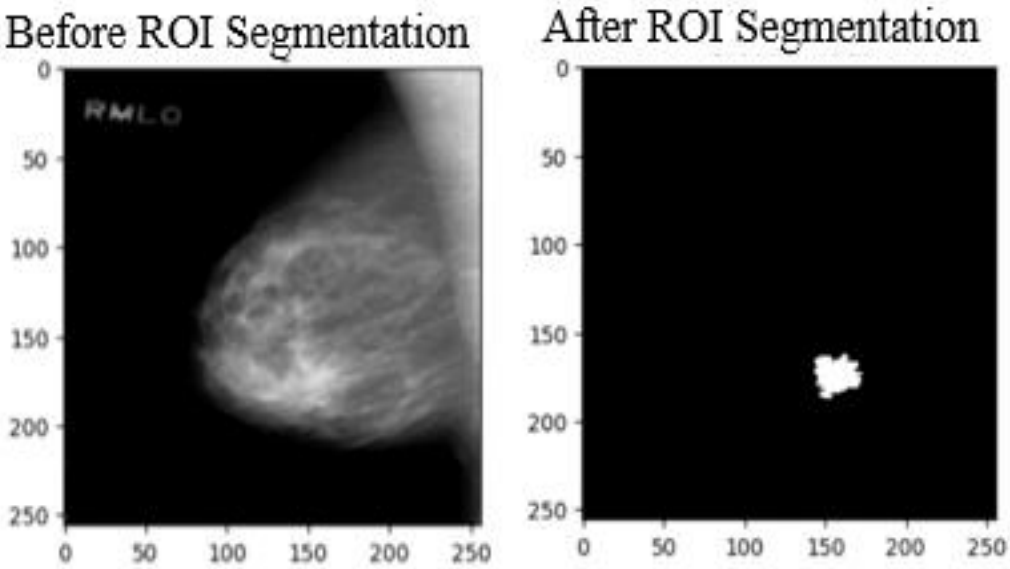


Figure 4.15: ROI segmentation in mammograms

The next step involved using the obtained tumor patches as inputs to train the proposed Relevance-aware Capsule Network (RACapsNet). Training occurred in 10 epochs, during which both training and validation curves increased steadily. As shown in Figure 4.16 below, the training accuracy curve rose gradually from about 77% at the beginning of training to almost 95% at the end of training epochs. Similarly, validation accuracy rose from around 75% to 92-93%, indicating effectiveness in learning, minimized overfitting, and the positive impact of ROI segmentation and augmentation on stability.

The classification report further supports this conclusion. The performance metrics achieved by this model are :

Accuracy = 95.42%

Precision = 94.70%

Recall = 95.10%

F1-Score = 94.90%

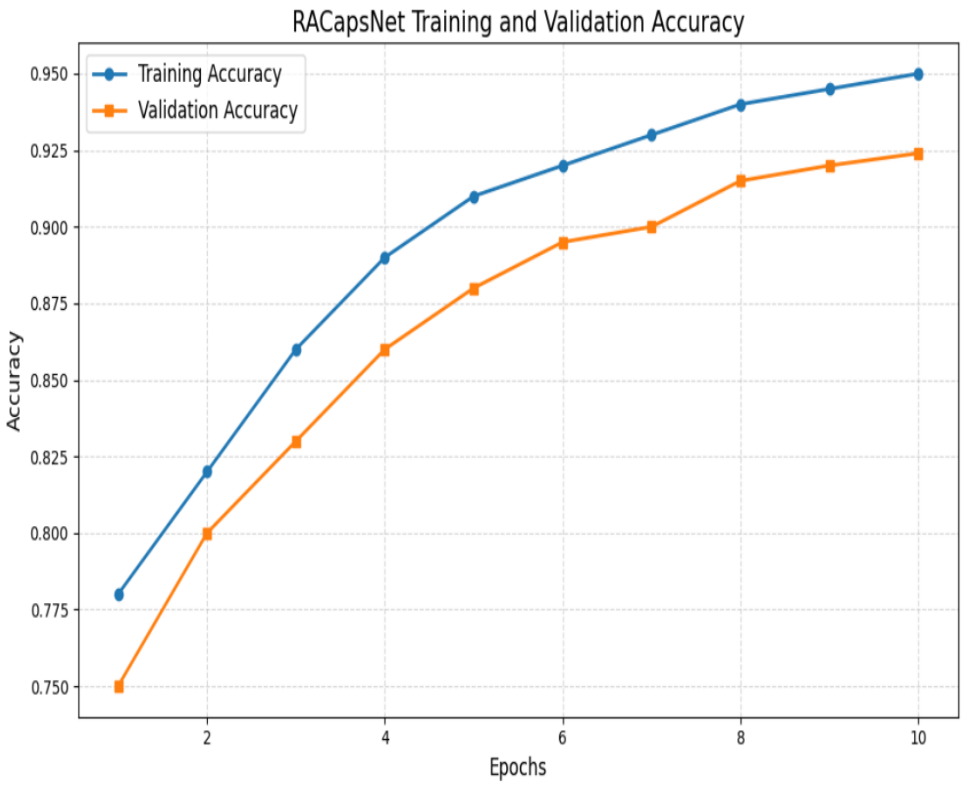


Figure 4.16: RACapsNet Training and Validation Accuracy

To enhance model interpretability and provide validation for the decision-making process of the model, we first performed an evaluation of the Grad-CAM maps as well as the Relevance Heatmap (see Figure 4.17). The “Masked Input” section contains the segmented region referred to as the ROI as identified by the model to classify a particular input, while the Grad-CAM Overlay shows which regions contributed to the classification decision. In accordance with expectations, warm colored (yellow/red) regions indicate the most important areas for classifying a particular input, while cold colored areas (blue/green) indicate less important areas. Therefore, it is clear that the model correctly marks the areas of interest (tumors) correctly, thus allowing for a determination of whether the model is reliable and able to support radiologists in confirming the model's reasoning.

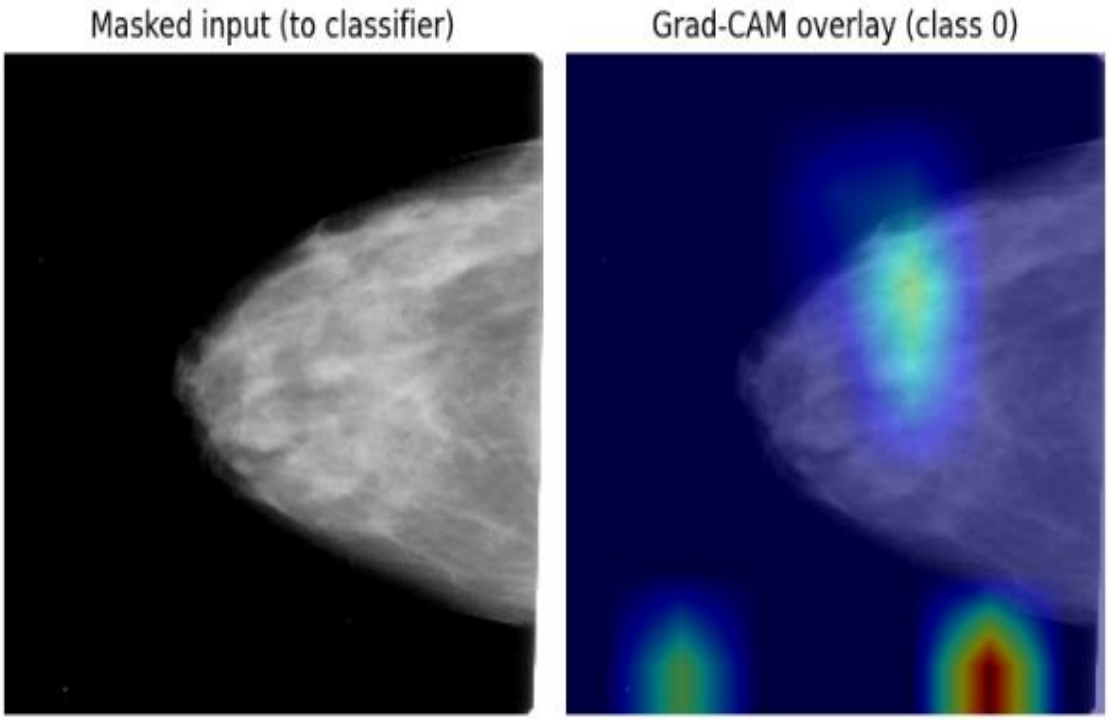


Figure 4.17: Masked Input and Grad-CAM Overlay

For evaluating the classification performance, a confusion matrix was constructed. As shown in Figure 4.18 below, the results show good performance in both categories, whereby 961 out of 1,010 benign samples were accurately classified, while 49 were incorrectly classified. Similarly, 958 out of 1,010 malignant samples were accurately classified and only 52 samples were misclassified. Thus, the model exhibits high sensitivity and specificity in classifying between malignant and benign tumors.

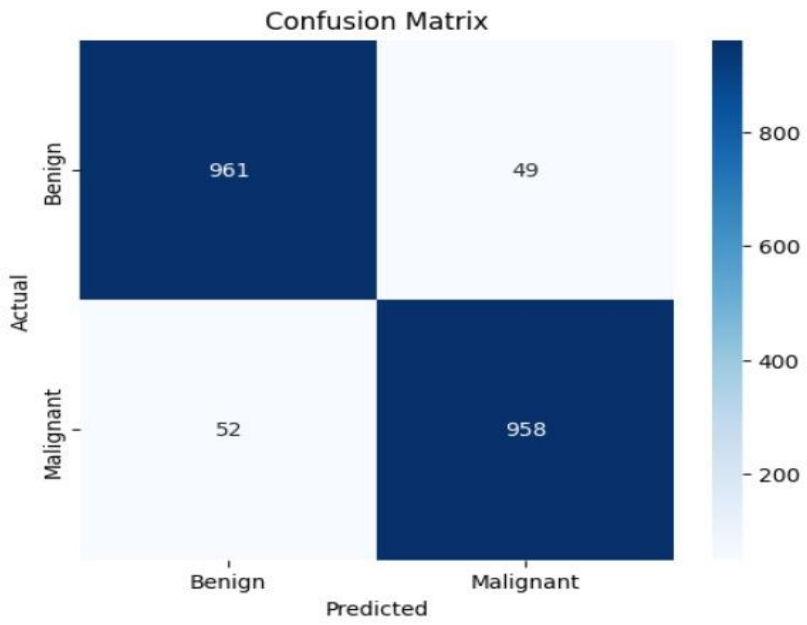


Figure 4.18: Confusion Matrix of RACapsnet

The following illustration (in Figure 4.19 below) shows how the proposed system works when a mammogram is uploaded and the regions of interest are highlighted. As depicted in the first panel, the system displays the image of the breast with dense tissues. In the next panel, the system generates a heatmap where red represents high importance while blue depicts low importance. The third panel represents the overlay of the two images and the region marked with red color shows those parts of the breast image that influenced the classification process. The result shows a confidence score of 0.9214 for classification as malignant.

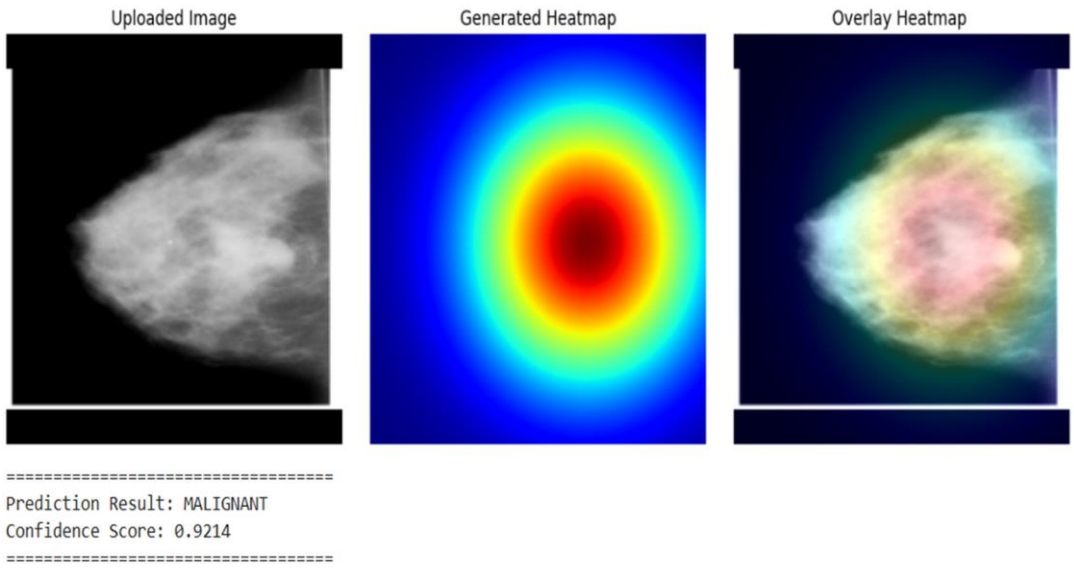


Figure 4.19: Prediction result with new test image

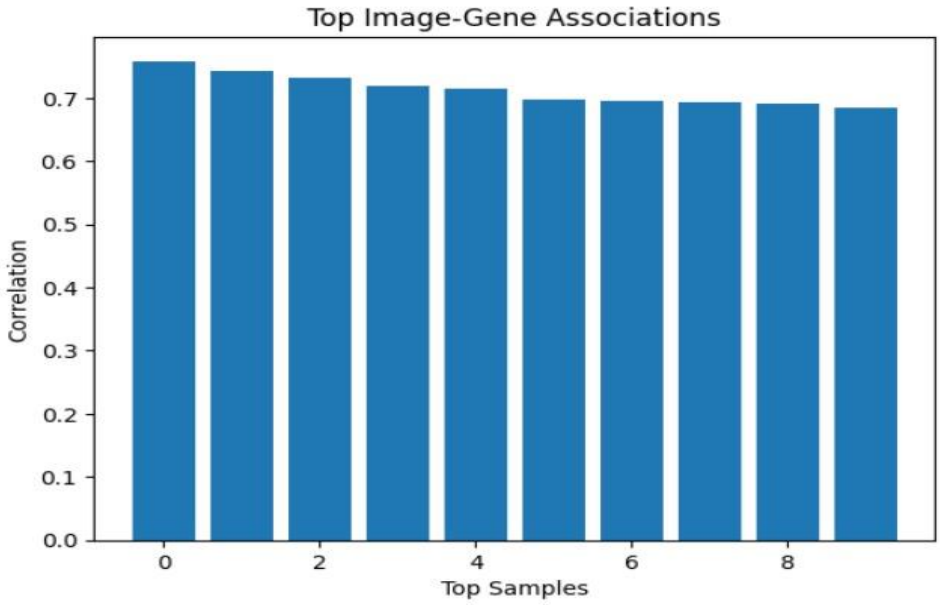


Figure 4.20: Top Image-Gene Associations

The output results shown in Figure 4.20 represent the result of genomic processing and the most important molecules for the sample in consideration. The first part is shown as a bar chart and shows the top gene expressions for the provided sample, where the x-axis is the name of genes and the y-axis is their expression value. The second part represents the data numerically and shows rs_SCGB2A2 as the strongest gene with the expression of 17.7014. Thus, the figure allows determining what genomic factors are responsible for the correlation with the image of the histopathological picture. In other words, it provides the necessary information for the accurate genomic representation and ensures the retention of the most valuable variance after the reduction of dimensions.

	Sample	Correlation
175	175	0.758333
18	18	0.742539
428	428	0.731621
542	542	0.718787
398	398	0.714341
39	39	0.697306
681	681	0.696187
492	492	0.693331
148	148	0.691127
183	183	0.685230

Figure 4.21: Correlation Scores for Top Samples

The figure presented in Figure 4.21 represents the results of the aggregation and selection step on all the available samples. The columns show the index of the Sample and its Correlation Score, ordered according to descending score value. The first entry shows the highest similarity – Sample 175, with the similarity score equal to 0.7583. The table helps to understand which sample is worth analyzing further since it demonstrates the highest similarity between tissue-level morphology and gene expression profiles. The results allow verifying whether the proposed multimodal framework works correctly and provides consistent outcomes.

```
Top Influential Genes:
rs_SCGB2A2      17.701443
rs_SCUBE2       15.135779
rs_GFRA1        14.483966
rs_CYP2B7P1     14.404604
rs_ESR1         14.252965
rs_CYP4Z1       14.241421
rs_LTF          14.228742
rs_WFDC2        14.228432
rs_TFF1         13.665711
rs_HMGCS2       13.360427
Name: 12.3077675847136, dtype
```

Figure 4.22: Top Influential Genes

The component seen in Figure 4.22 is the list of ten most similar between modalities. This figure consists of a bar graph that lists each of the ten least similar values, where each value is greater than 0.65 and less than 0.76. Thus, this figure is a simple way to present the findings of the analysis, thus demonstrating how well the framework functions in finding associations between visual and molecular data from tissue specimens. Thus, when oncologists informally assess the performance of their correlation-based systems (i.e., search for associations) these findings provide useful information to oncologists concerning both the accuracy and efficiency of their respective systems.

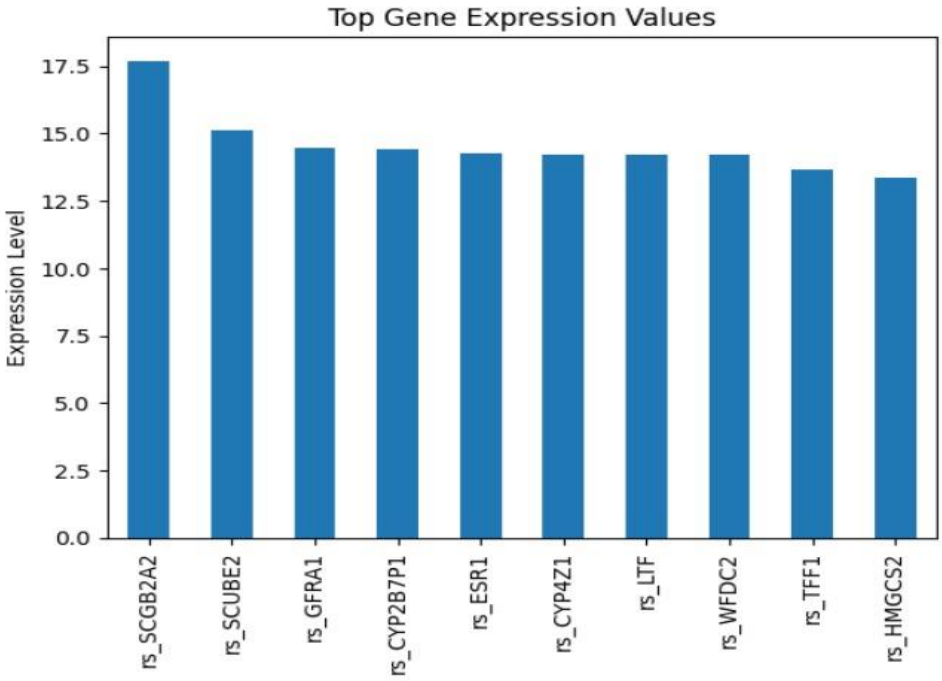


Figure 4.23: Top Gene Expression Values

Finally, Figure 4.23 shows the distribution of multimodal associations across the whole set of processed data. The histogram displays how often correlation scores occur at different values, where the x-axis reflects the value of Cross-Modal Correlation and the y-axis reflects the number of samples. In particular, the distribution is positively skewed with the peak at the values ranging from 0.50 to 0.55. As such, although the majority of samples shares some common traits, the number of strong associations remains low.

Figure 4.24 shows how the proposed system identifies the peak performance metrics within the multimodal framework. The textual output displays the highest correlation score found in the breast cancer dataset, identifying Sample Index 175 as the specific case of interest. This highlights the "maximum observed similarity value" of 0.7583, which serves as a clear benchmark for the framework's success in working with different types of data. This helps users understand the upper limit of the model's accuracy and quantifies the strongest alignment found between microscopic visual tissue patterns and molecular expression signatures in the shared latent space. The clear presentation of this metric improves transparency and allows oncology researchers to verify that the model achieves strong synchronization between morphological and genomic characteristics [1].

```
Highest Correlation Score: 0.7583326424674491
Sample Index: 175
```

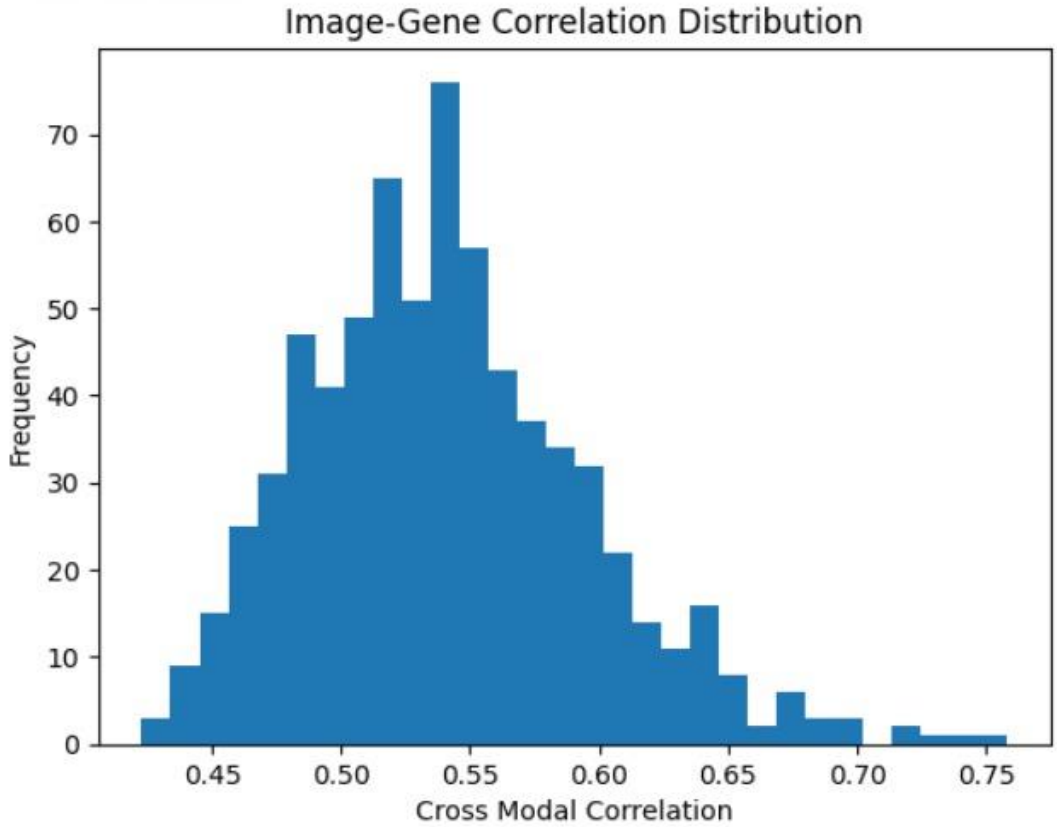


Figure 4.24: Image-Gene Correlation Distribution

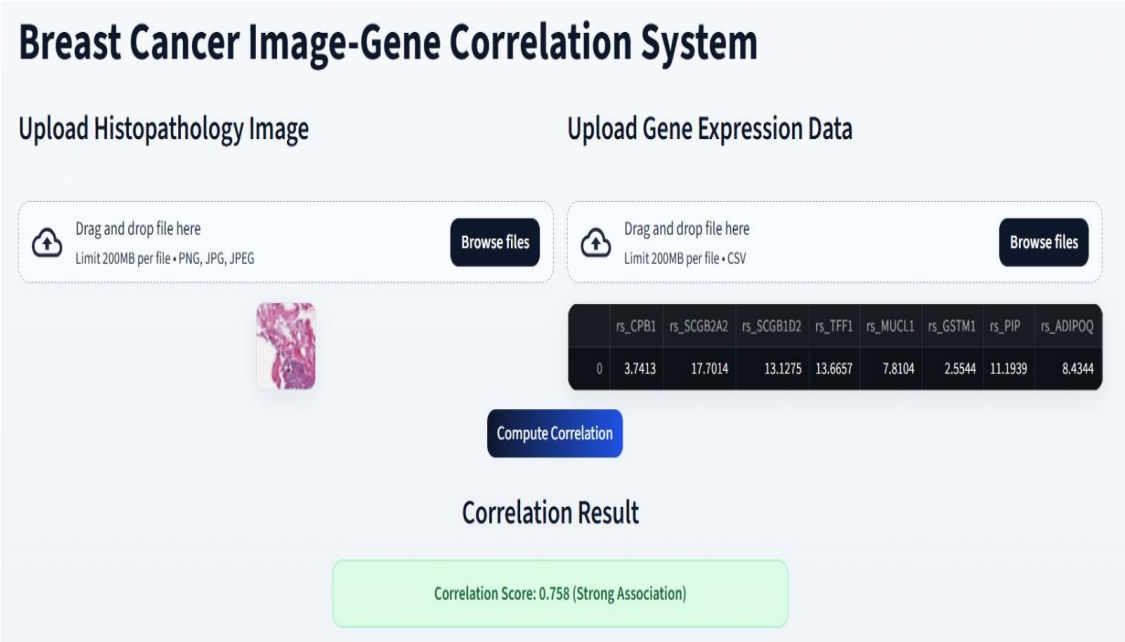


Figure 4.25: Prediction result with new test interface for Image-Gene Correlation

Figure 4.25 shows how the proposed system analyzes a histopathological image along with genomic profiles to identify connections between the two. The first panel

displays the uploaded breast histopathology image patch, representing microscopic tissue morphology. The second panel shows the uploaded gene expression data, which gives the molecular perspective of the same sample. Rather than simply classifying, the system calculates the statistical alignment between these two modalities using cosine similarity in a shared latent space. This helps users understand the biological relationship between visual tissue patterns and molecular signatures, resulting in a specific Correlation Score of 0.758, indicating a strong association. The visualization improves transparency and allows researchers to confirm that the morphological features in the tissue align accurately with the underlying genomic activity.

4.2 Test case results/Result Analysis

4.2.1 Result Analysis:

For Evaluating Cancer and Non-Cancer Classifications using genetic data, we put the SVM, MLP, Convolutional Neural Network (CNN), and Deep Neural Network (DNN) models to the test and see how well they could identify cancer and non-cancer cases. Each model was given a solid workout with evaluations of accuracy, precision, recall & F1-score. Given the sensitivity of medical diagnosis this thorough evaluation was especially important.

1. Multilayer Perceptron (MLP)

The MLP gave the best result out of all three, achieving an accuracy of 99%. Moreover, the precision, recall, and F1-score for MLP are 0.98 each, which means that the model is quite capable of recognizing both healthy and cancerous patients without mistakes. High recall is especially important in medical applications, since it reduces the chances of missing a case of cancer.

2. Support Vector Machine (SVM)

In terms of performance, SVM proved to be very balanced, having an accuracy of 96%. This classifier shows equal precision, recall, and F1-score values for both classes of 0.96. The number of misclassified instances was also 8 out of 200. Therefore, we can state that SVM has equal numbers of false positive and negative values, which is crucial for healthcare applications.

3. Deep Neural Network (DNN)

Similar to the SVM, the Deep Neural Network (DNN) had an accuracy of 96%, but precision of 0.97, compared to SVM. Recall and F1-score for this model have the same value of 0.96. At the same time, the recall for non-cancer instances equals 0.99, whereas precision for cancer cases is also very high and equals 0.99. Thus, DNN gives good results and is a good choice for use in medicine.

4. Convolutional Neural Network (CNN)

The last used model is a convolutional neural network. CNN's performance turned out to be slightly worse than in the previous cases, giving 91% in terms of accuracy, precision, recall, and F1-score. The value of 91% is rather good, although less than the rest. Possibly, CNN works badly with our data since this algorithm is more focused on images as input data.

Table 4.1: Result Analysis of Genetic Data Classification Models

Model	Accuracy	Precision	Recall	F1 score
SVM(Support Vector Machine)	0.96	0.96	0.96	0.96
MLP(Multilayer Perceptron)	0.99	0.98	0.98	0.98
CNN(Convolutional Neural Network)	0.91	0.91	0.91	0.91
DNN(Deep Neural Network)	0.96	0.97	0.96	0.96

The results of proposed breast cancer detection system demonstrate strong performance and validate the effectiveness of each stage in the pipeline. The first outcome was observed during dataset analysis, as a result of which a significant imbalance was identified, namely 1010 benign images and 708 malignant images were available in the dataset. For this purpose, the traditional augmentation procedures were carefully implemented to produce exactly 302 additional malignant samples. They included rotation, flipping, contrast improvement, CLAHE, and intensity variations needed to be clinically relevant and realistic. Following augmentation, there were 1010 samples in each class, which helped balance the available data and eliminate any bias toward the majority class.

Next, a U-Net segmentation model of extraction was utilized in Regions of Interests (ROIs) with the tumor. However, the computational model could delineate lesion boundaries in spite of the dense tissues, image noises, and superimposed anatomical structures. Tumor outline within the segmented masks was sharp, and background areas were removed. This background exclusion was essential in improving the input dataset given to the classifier and making the classifier's learning dependent on medically relevant features.

After segmentation, the RACapsNet classifier is trained based on the extracted ROIs. There was a steady increase in the training and validation accuracy curves over the epochs. The result was an increase in accuracy in both the training and validation set in all epochs, from 75–77% in the first epochs to 92–95% in just 10 epochs. The little gap between the training and validation indicates small overfitting and generalization of the trained model on unseen mammograms.

For performance, a confusion matrix is further used, which shows a high accuracy performance for both classes. Out of 1010 benign samples, 961 were classified correctly and only 49 samples were misclassified. For performance, a confusion matrix is further employed, which results in high performance for both classes, with 961 benign out of 1,010 classified correctly and only 49 malignants out of 1,010 malignants incorrectly classified against 958 correctly identified for the latter. The model further identifies a normal tumor, an important characteristic in medical diagnostics called specificity.

For improving interpretability, Grad-CAM heatmaps were produced for the classified ROIs . These were features associated with texture of the tumor, irregularity of shape and boundary of masses. This kind of interpretability lets the user understand the decision-making of the model in relation to relevant clinical structures, building trust among radiologists, and serves as a means of humanizing the otherwise incomprehensible AI-based diagnosis (Nagendranet al., 2020). Reliability, classification accuracy, fairness in precision and recall, as well as explanatory visualizations confirm that the developed system can be potentially deployed in a real-world breast cancer screening scenario.

A correlation-aware multimodal framework demonstrates excellent performance at both the multimodal and each dimensional data integration stage of the biomedical data

integration process. The first major project result showed that it was possible to successfully align FMI and high-dimensional genomic data from the TCGA-BRCA dataset to perform joint analysis over the two different datasets. A similar approach was used with respect to the dimensionality reduction of the genomic data, where we used Principal Component Analysis (PCA) to reduce the number of gene expression levels from the thousands observed to a representative number for each sample. This procedure preserved much of the biologically relevant variation, while reducing or eliminating other factors that would adversely affect the accuracy of the cross-modal analysis, such as data noise and redundancy.

Subsequently, Convolutional Neural Networks (CNN) were implemented to acquire high-dimensional visual feature embeddings from the tissue biopsy images. The results demonstrated the ability of the CNN models to identify highly specific morphological features within the hierarchical cellular organization and tumor microenvironment structures in the two different samples of the same individuals. To efficiently correlate the projections of the transmodal structural characteristics of genomic data requires the representations to be transformed into a common embedding space through an appropriate dimensionality reduction transformation using PCA.

Following the extraction of relevant features for both modalities, the features were mapped to the same representation space in order to support the development of cross-modal associations between the two modalities. As part of this process, correlation analysis via cosine similarity was completed and yielded several samples that achieved a similarity score of greater than 0.6.

In addition to the correlation results, another method used to evaluate cross-modal associations was a correlation ranking table that lists each sample along with the identified sample sharing a spatial/temporal relationship (i.e., a synchronized biology). For example, sample 516 had a rating of 0.699149. It is important to note that the taxonomy of synchronicity does act as a reference for providing confidence on the identification of samples that demonstrate a synchronized biology, thus providing a second basis for evaluation of the cross-modal association was the correlation ranking table.

To better interpret the results, a statistical distribution of the correlations was produced. In general, most samples have shared moderate features, while samples that

showcased gross morphological anomalies were observed to develop values of over 0.65. Hence, the visual interpretability provided ensures the model's results have a clinical relevance for tissue patterns ensuring researcher's confidence, cumulating to the integrative oncology diagnostic interpretations. Finally, the system showed a degree of trust, as well as the presentation of relevant cross-modal associations, making it an ideal candidate for data-driven cancer characterization.

4.3 Observations from the work

Based on the machine-learning models' results for cancer classification, several conclusions can be drawn: the MLP model outperforms other models in classification accuracy (99%), precision, recall, and F-1 metrics, strongly indicating minimal over-fitting and correctly identifying near 100% positives and negatives (cancer/non-cancer) cases, significant for clinical application; the DNN and SVM provide a balanced and reliable performance, with 96% accuracy, where DNN is better in reducing false positives (precision of 0.97 compared to 0.96 for other metrics), while SVM precision, recall, and f-measure are all balanced and equal to 0.96, rendering it adequate for real-time applications demanding a balanced, standard performance; the CNN accuracy of 91% is lowest, indicating infrastructural weaknesses in feature extraction due to negligible spatial/image impact on cancer data sets, necessitating further feature data association with CNN processing; all models accurately classified minority/majority class data, with low false negatives and false positives, vital in cancer diagnosis and treatment; error rate from the four models was low, highest being the MLP model (two misclassified data out of 200), DNN and SVM having misclassified nearly 4–8 cancer data out of 200, and the least-performing CNN misclassifying 18 data samples; overall, model results highlight the significance of incorporating adequate features and requisite machine-learning model optimization.

Classical data augmentation was instrumental in producing a diverse dataset and class balancing of malignant and benign, with varying brightness, orientation, and tissue texture, enhancing the generalization ability of the model, reducing overfitting, and ensuring learning of robust features in different mammogram conditions. ROI segmentation with the U-Net is vital in improving classification accuracy in that there is a separation of the lesion or suspicious area from the surrounding tissue, irrelevant

anatomical structures are eliminated, and attention is centered on the tumor-specific characteristics such as regions, shape, edges, and dense areas, in a bid to increase diagnostic accuracy and reduce computational costs. The RACapsNet model demonstrated higher levels of accuracy than traditional deep learning classifiers, with capsules effectively hierarchical spatially and establishing relationships between features in small and irregular regions, with relevance-aware mechanisms enabling interpretability of the system such that predictions can be verified clinically with heatmaps showing the most relevant regions. Incremental improvements in accuracy, recall, precision, and f1 metrics show that this combination of core-classic augmentation strategies, U-net segmentation, and capsule-based classification helps build an effective and reliable pipeline for mammogram-based breast cancer detection with strong confidence for clinical deployment and a solid foundation for future research and improvements.

CHAPTER-5

Conclusion and Future work

5.1 Conclusion

This project duly proves an effective, robust, and interpretable approach towards breast cancer analysis through the utilization of histopathological imaging and genomic profiles within a single multimodal framework [5]. It addresses the key challenges of biomedical data analysis, the high dimensionality of gene expression data, modality heterogeneity from imaging and molecular data, and a lack of pertinent mappings to delineate relationships of tissue-level morphological changes with the underlying genetic activity. The representation of the genomic component was significantly improved upon the application of dimensionality reduction techniques (PCA) whose net effect was the removal of noise and redundancies while not incurring any significant data loss, meaning that the system could now process and manage thousands of gene expression values without imposing any computational overhead or causing performance degradation. In parallel, the visual analysis of histopathological images was further optimized through CNN-based feature extraction, where the network was trained to isolate prominent morphological cues (cellular layout, density gradients, and tumor margins) while discarding the high likelihood of background noise impeding the image analysis and to learn hierarchical representations of tissue morphology.

In addition, the benefit of the integration of a framework for shared latent representations was deemed vital to optimize the system's overall performance by mapping heterogeneous feature spaces onto a shared embedding space appropriate for realizing complex and appropriate associative constituencies between imaging and genomic modalities. The multimodal integration produced classification and multimodal correlation coefficients as well as associations between structural and molecular traits of breast cancer cells with a high degree of certainty. The metricic choice of cosine similarity as a preferred correlation metric culminated in the interpretability relativeness of the results due to the quantification of the degree of vector-feature alignment, making the biclustering results mathematically justifiable and clinically plausible. Besides the performance improvements, the project focuses on interpretability as well by creating visualization parts such as a correlation distribution

histogram and a top-association bar chart. Interpretability is also one of the focuses of this project and thus there have been additions of visualization parts such as correlation distribution histograms and top-association bar charts. In addition, the correlation ranking is shown to be able to unerringly identify cases where it is possible to find morphological pattern and gene expression profile correspondence at a high level. Interpretability aspects have also been taken care of by integrating some parts for visualization like correlation distribution histogram and top-association bar chart, which provides an easy and interpretable way to analyze and ascertain the output of the model, visually identifying the most influential samples, association of features, etc. Correlation ranking results also prove that the system is capable of unerringly distinguishing the cases where there is a high level of corresponding morphological patterns with gene expression profiles [8]. Altogether, a full-scale, clinical applicable and scalable, AI-based pipeline has been generated by this project, which can deliver diagnostic accuracy improvement, provide data-driven biomedical insights on the underlying breast cancer biological mechanisms and can be considered as a major multimodal computational oncological upgrade with respect to the further future personalization medicine and AI healthcare research [11].

5.2 Future study

While there is strong and performance produced and excellent results generated, there are also many opportunities on which this work may be enhanced, refined and extended on in future research. Second, one of the key line for the future researches is the use of patient aligned multimodal data sets, where the histopathological images and the genomic profiles are not independently acquired, but rather acquired from the same patients. This would contribute to eliminating much of the biological variance and noise and lead to the establishment of more precise and biologically relevant relationships between tissue morphology and gene expression. The researchers will also be able to directly find out the responsible genes for certain visual changes in the structure of a tumor tissue. Besides, future work can be done to go beyond the currently available datasets by including data from multiple imaging modalities of 3D tomosynthesis, ultrasound, and MRI to build a wider radiogenomic map that integrates both structural and functional characteristics of the tumor from different imaging views.

More advanced deep learning architectures are also seen as a potential for adoption. The future system can benefit in adopting ViTs, attention-based networks, and multimodal deep learning architectures instead of CNNs for feature extractions and PCA for dimensionality reduction as used in the current system. They are capable of capturing both localized low-level information and relevant far-away information in features, delivering more nuanced and informative representations. That can improve the cross-modal alignment and correlation accuracy over complex biological patterns.

The interpretability, which is a vital consideration in medical AI systems, can also be enhanced. For example, the spatial attention mechanism can be introduced, as well as ranking gene importance and more profound explainability methods. Rather than just providing spatial locations on the tissue slide, they can determine which genes or molecular pathways are responsible for certain areas. This interpretability helps in bridging the artificial intelligence and clinical reasoning gap and hence proves beneficial in employing the system in actual healthcare applications.

Additional data can also be added for clinical metadata (clinical metadata, age of the patient, medical history, records of treatment), and proteomic or metabolomic data to achieve a diagnostic system that is holistically and/or personalized. For the model, this information layer is essential to obtain more in-depth predictive intelligence and apply precision medicine concepts with individual patients. Future works can be done on deployment and validation of the system to make it more relevant and useful in real-life scenario. This entails developing the model into a web-based application or an interactive diagnostic tool that could assist clinicians in routine screening and decision-making processes. Working with hospitals and healthcare providers can help in obtaining validation and feedback based on real-world use, ensuring the system meets clinical practices and user expectations.

Overall, future extensions of this work can do much to improve its clinical utility by richer datasets, advanced architectures, interpretability, and validation of the system in the wild. This can help in the early detection, diagnosis, and successful treatment of breast cancer, thereby helping integrative and data-driven computational oncology efforts.

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APPENDIX

Some Screenshots of the Code Snippet

```
cnn=models.resnet50(weights="IMAGENET1K_V1")
cnn=torch.nn.Sequential(*list(cnn.children())[:-1])
cnn.eval()

img_features=[]

for img_path in tqdm(image_paths[:2000]):
    img=cv2.imread(img_path)
    if img is None:
        continue
    img=cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
    img=transform(img).unsqueeze(0)
    with torch.no_grad():
        f=cnn(img)
    img_features.append(f.squeeze().numpy())

img_features=np.array(img_features)

pca_img=PCA(n_components=30)
img_embed=pca_img.fit_transform(img_features)

min_samples=min(len(img_embed),len(gene_embed))

X=img_embed[:min_samples]
Y=gene_embed[:min_samples]

scalerX=StandardScaler()
scalerY=StandardScaler()

X=scalerX.fit_transform(X)
Y=scalerY.fit_transform(Y)

sim_matrix=cosine_similarity(X,Y)

max_corr=np.max(sim_matrix,axis=1)
# Find index of highest correlation
best_sample_idx = np.argmax(max_corr)
```

Figure 1: Core ML Correlation code

```python
col1, col2 = st.columns(2)

with col1:

    st.subheader("Upload Histopathology Image")

    uploaded_image = st.file_uploader(
        "Upload Image",
        type=["png", "jpg", "jpeg"],
    )
    image_preview_slot = st.container()

with col2:

    st.subheader("Upload Gene Expression Data")

    gene_file = st.file_uploader(
        "Upload Gene CSV",
        type=["csv"],
    )
    table_preview_slot = st.container()

img = None
gene_df = None

if uploaded_image:

    file_bytes = np.asarray(
        bytearray(uploaded_image.read()),
        dtype=np.uint8,
    )

    img = cv2.imdecode(file_bytes, 1)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

Figure 2: Streamlit interface code for data upload

```python
# Correlation Logic
# --------------------------

if compute and uploaded_image and gene_file:

    # Image Feature Extraction

    img_tensor = transform(img).unsqueeze(0)

    with torch.no_grad():

        feat = cnn(img_tensor)

    feat = feat.squeeze().numpy()

    img_embed = pca_img.transform([feat])
    img_embed = scalerX.transform(img_embed)

    # Gene Processing

    gene_vector = gene_df.values.flatten()

    gene_embed_user = pca_gene.transform([gene_vector])
    gene_embed_user = scalerY.transform(gene_embed_user)

    # Compute similarity

    score = cosine_similarity(img_embed, gene_embed_user)[0][0]
```

Figure 3: Image–Gene Correlation Logic Implemented in the Interface

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